

Merchants of Death: The Effect of Credit Supply Shocks on Hospital Outcomes[†]

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This study examines the link between credit supply and hospital health outcomes. We use bank stress tests as exogenous shocks to credit access for hospitals that have lending relationships with tested banks. We find that affected hospitals shift their operations to increase resource utilization following a negative credit shock but reduce the quality of their care to patients across a variety of measures, including a significant increase in risk-adjusted readmission and mortality rates. The results indicate that access to credit can affect the quality of health care hospitals deliver, pointing to important spillover effects of credit market frictions on health outcomes. (JEL G21, G32, I11, I18)

Hospitals play an essential role in maintaining public health. Hospitals are also crucial to the economy, with health care spending in the United States accounting for 18–20 percent of GDP and the hospital sector accounting for one-third of this spending.¹ However, like other enterprises, hospitals must obtain financing for their operations and utilize credit markets for this financing. Indeed, the vast majority of US hospitals carry leverage and often rely on debt for their financing needs. Moreover, due to low profit margins, hospitals carry considerable default risk, with health care defaults on municipal bonds comprising 20 percent of all bond defaults, second only to housing. Consequently, frictions in the credit market can exacerbate impediments to credit access for hospitals, straining hospital finances and potentially influencing real decisions and outcomes.

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¹US health care spending was \$4.1 trillion in 2020, constituting 19.7 percent of US GDP. Furthermore, hospital employment in the United States exceeds 7.05 million, and hospitals are among the top employers across US cities (Laughlin et al. 2021). See also Gaynor, Ho, and Town (2015). Moreover, the economic decline following the 2020 pandemic is reported to be partly attributed to the large reduction in health care spending, leading to significant layoffs of hospital medical staff. See “Plunge in health-care spending a big reason US economy sank in first quarter,” *CNBC*, April 29, 2020. Health care spending in other OECD countries is similar, with an average of 8–9 percent of GDP.

Given the prevalence of borrowing in the hospital sector (among all types of hospitals, including nonprofit and rural hospitals), it is important to understand the role of credit access in shaping hospital operating decisions, as well as the potential spillover effects of such decisions on the hospital's quality of care. The questions we seek to address are thus: (i) how do shocks to credit markets transmit to hospital finances? (ii) how do hospitals respond to negative shocks to credit access in terms of financing and operating decisions? and finally (iii) do we observe indirect, negative effects on patient health outcomes following tightened credit constraints? Given their importance to public health, we would expect (or hope) that hospitals can maintain the same quality of care despite frictions in financial markets. This question highlights an important yet overlooked negative societal externality (health consequences) that can arise from credit shocks. Research on this topic may therefore have important social consequences and policy implications.

To help shed light on the above questions, we utilize the staggered pattern of stress tests on US banks implemented by the 2010 Dodd-Frank Act to cleanly test the effects of shocks to the supply of credit. Stress tests are regulatory assessments by the Federal Reserve designed to gauge a bank's ability to withstand an impending economic crisis, the first of which were implemented in 2012. Following a stress test, banks often engage in risk management actions in order to improve their solvency and capital adequacy ratios. We use the fact that a given hospital's bank experiences a stress test as an exogenous negative shock to credit for the hospital. As noted by Gao, Lee, and Murphy (2019), hospitals are particularly risky borrowers, with higher than average yields and default rates for municipal bonds. Consequently, to better manage their risk or improve their capital adequacy, stress-tested banks can lower the amount of credit provided or demand higher rates from these risky borrowers (Acharya, Berger, and Roman 2018; Cortés et al. 2020).²

Using a staggered difference-in-differences specification over the period 2010–2016, we examine the change in operating decisions and performance, as measured by patient health outcomes, between hospitals subject to a credit supply shock—hospitals that had lending relationships with banks which were later stress-tested—relative to hospitals which did not experience a shock. This empirical strategy has the advantage that (i) the stress tests themselves are unrelated to the underlying health of a local population; (ii) the tests occurred in a staggered manner; (iii) the tests were applied to banks based on size thresholds rather than on bank performance; and (iv) it is unlikely that hospitals could anticipate the negative bank responses following a stress test.

We first establish that bank stress tests constitute a negative credit shock to their connected hospital borrowers. In particular, we find that loan spreads increase while loan amounts and maturity periods decrease for affected hospitals, and these hospitals are more likely to switch lenders to one for which they did not have a previous relationship with.³ These results are consistent with bank stress tests increasing the

² As shown by Acharya, Berger, and Roman (2018) and Cortés et al. (2020), banks trim their loan portfolios and charge higher rates for riskier loans following a stress test, thus constituting a negative credit supply shock to firms that borrow from these tested banks.

³ Changing lenders or acquiring loans from new banks also proves problematic for hospitals, as new lenders require a higher rate to compensate for the more severe information asymmetry due to the absence of a previous relationship.

cost of credit for an affected bank's hospitals and reinforce the findings of Acharya, Berger, and Roman (2018) and Cortés et al. (2020). The increases in debt servicing costs are economically significant for hospitals in our sample, due to low profit margins within the hospital industry. For example, the impact of the higher spread (interest rate over LIBOR, the London interbank offered rate) alone results in an additional \$1.07 million in interest expense per year for an affected hospital in our sample, which amounts to an economically sizable 13.4 percent of net income. A decrease in net income heightens the likelihood of hospital insolvency and closure—an ever-present risk in the hospital industry (Bai et al. 2020)—and also leaves the hospital with fewer internally generated funds with which to finance future operations.

We then explore how hospital financial and operating outcomes change as a result. We find that, in response to the credit shock, affected hospitals improve profit margins on services and ramp up internally generated funds (i.e., net income). In particular, in response to tighter credit conditions, we find evidence that hospitals rely more on their existing resources by increasing bed utilization and discharge rates. In a given year, each bed in an affected hospital is occupied by eight more days per year, on average, relative to an unaffected hospital. Moreover, affected hospitals accommodate 2.4 more patients per bed each year, which amounts to 191 additional patients accommodated per year for the average affected hospital. These utilization effects are consistent with prior literature that has documented an increase in efficiency following stricter financial constraints (Hovakimian 2011).

While the previous results suggest that hospitals work to improve their financial efficiency through expanding their profitable operations in response to tightening credit, we find this comes at the expense of health care quality for patients. More specifically, affected hospitals experience a significant *decline* in quality of care and patient health outcomes. We examine patient health outcomes following treatment using risk-adjusted, unplanned 30-day hospital readmission rates for various health conditions, a widely used measure by both government agencies and academic researchers for quality of care that assesses the effectiveness of treatment.⁴ We also gather data on 30-day risk-adjusted mortality rates for similar conditions. Finally, as a direct measure of patient satisfaction with the quality of care and attentiveness, we utilize patient survey data. This data includes patient satisfaction following discharge regarding hospital quality, communication with physicians and nurses, efficacy of pain control, and other items relevant to the treatment and hospital stay.

Across all sets of measures, the results show hospital performance declines following credit supply shocks. We find that patients discharged from affected hospitals are significantly more likely to be readmitted within 30 days from discharge. This result is strikingly consistent across the three diagnostic groups for which we have detailed data (heart failure, acute myocardial infarction, and pneumonia) and also holds for a wider set of medical conditions. The magnitude of the effect is sizable.

⁴For example, the Centers for Medicare and Medicaid Services, under the Hospital Readmissions Reduction Program, uses unplanned readmission rates as the central performance criteria when determining Medicare payment reductions. Moreover, rehospitalization accounts for more than \$17 billion in avoidable Medicare expenditures and is associated with poor outcomes (Jencks, Williams, and Coleman 2009). A substantial portion of readmissions are estimated to be preventable (MedPAC 2007). See also <https://www.cms.gov/Medicare/Medicare-Fee-for-Service-Payment/AcuteInpatientPPS/Readmissions-Reduction-Program>.

Restrictions to the access of credit for hospitals leads indirectly to an additional 1,570 patients readmitted per year in aggregate across affected hospitals. Similarly, with respect to patient mortality from pneumonia—a common hospital-acquired condition (Rothberg et al. 2014)—the results show an increase of 718 patient deaths per year across affected hospitals.

With respect to patient experiences, patient evaluations regarding efficacy of treatment and attentiveness of the medical staff are consistently lower for affected hospitals. Across *all* eight rating dimensions, recently discharged patients from affected hospitals are significantly less satisfied, including with regard to lower overall care quality, less communication with doctors and nurses, and worse pain control. Collectively, these results suggest that patient health outcomes and quality of care are adversely affected for hospitals which experience a shock to credit access.

We further utilize patient-level data for several states to better understand the mechanisms underlying the above-mentioned results. We find that affected hospitals increase their income and profit margins through increased resource utilization and cost efficiency. Specifically, affected hospitals increase admissions per diagnosis-related group (DRG) by 4.6 percent, in line with our earlier results that discharges and bed utilization increase for affected hospitals. Moreover, the increase in admissions is driven by greater inpatient admissions from patients who arrive at the emergency department (ED) and through an increase in scheduled nonemergency (i.e., elective) procedures which require an inpatient stay (such as hip replacements). Interestingly, inpatient admissions for DRGs with *lower* relative weights (which reflect lower complexity or severity of the patient's condition) see a *larger* increase. These findings are consistent with hospitals lowering the threshold for inpatient admission and admitting more patients with less severe conditions. These results also comport with anecdotal evidence and Department of Justice cases whereby hospitals use higher inpatient admissions to increase revenues (Pomorski 2021).

In terms of cost efficiency, affected hospitals significantly reduce the number of procedures provided for admitted inpatients across all insurance types and also reduce inpatient length of stays (for privately insured patients). Such actions lower costs for hospitals while maintaining similar payments for inpatient admissions reimbursed on a fixed payment scheme. Collectively, these responses illustrate how hospitals increase revenues while also lowering costs to improve profitability.

To shed light on the decrease in care quality, we examine hospital staffing using the patient-level data mentioned above. We find that physicians are on average providing care for a greater number of patients per DRG and that the unique number of physicians providing care per DRG remains the same at affected hospitals. These results suggest that physicians face greater strains on their time and attention following the credit supply shock. We additionally utilize data on hospitals' use of timely and effective treatment and procedures by medical staff for certain medical conditions to measure attentiveness and care quality. As an example, this includes the frequency with which patients suffering from a heart attack received a percutaneous coronary intervention (PCI) within 90 minutes of arrival. The results show that affected hospitals exhibit increased delay in providing critical treatment and a lower propensity in performing requisite medical procedures for the specific medical conditions. For affected hospitals, the likelihood of failing to provide proper treatment for five out of six quality metrics increases by 0.5 to 1.3 percentage points,

which represents a 14 to 20 percent increase relative to the sample mean of 3.2 to 6.5 percent, depending on the treatment or procedure. This decline in the process of care scores is consistent with medical staff being less attentive to patients in affected hospitals, which helps to explain the increase in readmission rates reported above.

Taken together, the above findings imply that affected hospitals adjust for the increased cost of debt or the decline in external financing by increasing cost efficiency and internally generated funds from patients. This includes greater admission of patients with less severe conditions and of privately-insured patients. The increase in inpatient admissions, however, is not met with a contemporaneous increase in hospital physicians, resulting in physicians providing care for a larger number of patients. Consequently, the heavier inpatient volume comes at the cost of worse performance: medical staff appear to be less attentive to patients, as evidenced by a decrease in the quality and timeliness of care, and patient health outcomes decline, as unplanned readmissions rise. In sum, hospitals attempt to “make up the difference” through higher resource utilization, but sacrifice quality of care in the process, which in turn results in worse health outcomes.

A question which arises from these results is whether the change in hospital operations implies that affected hospitals were operating suboptimally prior to the credit shock. Hospitals aim to maximize profitability, but, unlike other firms, hospitals also have a health provision objective (i.e., concerns for patient utility) that can run counter to profitability. Consequently, hospitals optimize between profits and health provision in their objective function. Our results imply that the tightening of financial constraints can lead hospitals to reoptimize and shift their decisions more towards profitability and away from health care quality.⁵

In additional analyses, we explore differential responses based on hospital characteristics as well as heterogeneous exposure to the treatment. As discussed above, we predict that the primary channel by which hospital performance declines is through frictions in credit access. Accordingly, under the hypothesized channel, hospital borrowers that are more affected by tightened credit constraints should experience a more pronounced effect in outcomes and performance. We find that the effects are stronger for hospitals that have a greater reliance on bank loan financing and smaller cash holdings prior to the stress tests. In terms of heterogeneous exposure to the treatment, we build from the observation that banks whose stress test outcomes are closer to the regulatory minimum have a stronger incentive to manage risk relative to banks whose projected outcomes are farther from the threshold (Cortés et al. 2020). This shorter distance from the threshold translates to a more severe credit supply shock for a bank’s corresponding hospital borrowers (e.g., through a greater reduction in lending or higher interest rates). We find that hospitals that borrow from banks whose outcomes are closer to the regulatory minimum exhibit stronger responses to the tightened credit constraints.

Finally, we consider a variety of robustness tests. These include running our results on a propensity score-matched sample and controlling for time-varying geographical differences. In addition, we explore robustness related to our sample composition, including samples restricted to hospitals that are in small hospital systems,

⁵Hospitals may also need to increase internally generated funds to prevent closure, a prevalent concern among many US hospitals (see, Capps, Dranove, and Lindrooth 2010; Bai et al. 2020; Pomorski 2021).

have for-profit status, or have borrowed from commercial banks. We also examine a placebo test centered on nonexposed rival hospitals.

This study relates to several different areas. Our paper contributes to the literature that examines the impact of financial frictions. This includes studies that document a negative impact on investment in the presence of constraints to credit access (Chava and Roberts 2008; Campello, Graham, and Harvey 2010; Duchin, Ozbas, and Sensoy 2010; Lemmon and Roberts 2010). The current study shows that shocks to credit supply can influence distinct firm decisions aside from investment, such as more granular firm operating activities. Moreover, our results indicate that such decisions can (indirectly) have real effects on health outcomes. As such, our paper ties into the strand of literature that studies the real effects of credit supply shocks (Gan 2007; Hombert and Matray 2017). Our study identifies a novel and important real effect (health consequences) arising from frictions in financial markets. Relatedly, our results show unintended downstream consequences of public policy decisions regarding the financial sector. This contributes to our understanding of how changes in public policy can affect bank lending activities and the potential spillover effects (e.g., Bernanke and Gertler 1995). The current study is also related to the large literature that studies relationship lending (Petersen and Rajan 1994; Boot 2000; Detragiache, Tressel, and Gupta 2008). We contribute to this literature by showing that a negative shock to relationship lending which reduces credit supply in turn reduces the *quality* of service of an important public good (health care). As a result, we provide novel evidence of how credit markets can indirectly affect health outcomes.

Our analysis is also related to the literature at the intersection of health care and finance. Adelino, Lewellen, and Sundaram (2015) use nonprofit hospitals to test the investment cash-flow sensitivity of nonprofit firms and find that these hospitals respond to increases in their cash flows (due to financial investments) by increasing their investments, in a similar way as public firms. Dranove, Garthwaite, and Ody (2017) and Adelino, Lewellen, and McCartney (2022) examine hospital responses following a drop in investments due to the 2008 financial crisis. Dranove, Garthwaite, and Ody (2017) find that the average nonprofit hospital did not respond to the crisis with price increases, but reduced unprofitable service offerings (with the reverse holding for nonprofit hospitals with greater bargaining power). Adelino, Lewellen, and McCartney (2022) find no aggregate evidence of a shift towards more profitable procedures due to the financial crisis, except by the most severely affected hospitals.⁶ Gupta et al. (2024) examine the effect of private equity investments in

⁶Both Dranove, Garthwaite, and Ody (2017) and Adelino, Lewellen, and McCartney (2022) consider a shock to hospital endowments (i.e., a wealth shock), while the present study examines a shock to operating profitability (i.e., an income shock) and access to external financing. The nature of the shock (wealth versus income) can play a critical role in how hospitals respond, as changes in endowment values may not translate to changes in distributions or income from the endowment. In particular, as we show, a direct shock to operating profitability in the form of higher interest expenses results in a more immediate response, as hospitals must cut costs and boost revenues to mitigate the cost increase, while sacrificing the delivery of health care quality in return. Additionally, there are several reasons why hospitals may not respond to financial shocks with price increases, as documented in Dranove, Garthwaite, and Ody (2017). For example, renegotiating prices with insurers may not be possible in the short-run if contracted prices with insurers have been set well in advance. Moreover, the hospital's ability to renegotiate prices depends on a number of factors that can be outside of the hospital's control, such as the market power of the hospital and insurer. Consequently, it can be more difficult for hospitals to adjust prices than to engage in other operational adjustments for which hospitals have more control over.

the quality of care delivered by nursing homes. Another stream of research investigates the impact of government health care reforms, such as the Affordable Care Act (ACA), on equity and debt prices. Koijen, Philipson, and Uhlig (2016) consider medical innovation and R&D, and document a premium in the equity returns of health care firms (including drug and biologic companies) due to the risk of government reforms. Gao, Lee, and Murphy (2022) examine the effect of the ACA on nonprofit hospital municipal bond spreads. Our paper contributes to the finance and health care literature by documenting a link between hospitals and credit markets and shows how credit markets can indirectly affect health care. To the best of our knowledge, the present study is the first to document the impact of credit access on patient health outcomes, quality of care, and patient satisfaction as indirectly arising from frictions in the credit market.

Our study is also related to the literature which considers potential inefficiencies in the health care sector. Prior studies have documented variation in treatment rates across providers (e.g., Fisher et al. 2003; Abaluck et al. 2016; Chandra and Staiger 2020; Einav, Finkelstein, and Mahoney 2023; see Chandra, Cutler, and Song 2011 for a review). We document an important substitution effect that may contribute to the observed heterogeneity: in the presence of (heterogeneous) financing constraints, hospitals turn to generating greater revenues through inpatient admissions, while reducing the number of procedures and average length of stay for inpatients. Furthermore, our paper is related to the literature on hospital cost-shifting, which considers potential increases in prices for private payers following reductions in public payments (e.g., Dranove 1988; Zwanziger and Bamezai 2006; see Frakt 2011 for a review). However, few studies in this literature consider *cost-cutting* by hospitals (exceptions include Cutler 1998 and Dranove, Garthwaite, and Ody 2017). We contribute to this literature by examining specific cost-cutting and revenue-increasing decisions, such as inpatient admissions and number of procedures, which, to the best of our knowledge, have not been considered in the extant literature. Moreover, we show that quality of care and patient health outcomes decline following the observed shift in utilization. Finally, our setting permits a research design with tight identification, allowing for a causal analysis.

The remainder of this paper is organized as follows. In Section I, we describe our institutional setting and conceptual framework in detail. In Section II, we discuss our empirical strategy and data. Section III presents the main results, while Section IV explores mechanisms underlying the main results. We discuss heterogeneity and robustness tests in Section V, and the final section concludes. Additional analysis is provided in the online Appendix.

I. Institutional Setting and Conceptual Framework

Stress Tests

Following the 2008 financial crisis, sweeping reforms regarding the regulation and monitoring of financial institutions were enacted through the Dodd-Frank Wall Street and Consumer Protection Act (DFA) of 2010. Among the reforms, Section 165(i)(2) of the DFA requires large bank holding companies (hereafter “banks”) to undergo

annual stress tests generated by the Federal Reserve under each of three scenarios (baseline, adverse, and severely adverse).⁷ The stress tests are intended to provide information about an individual bank company's ability to withstand potential economic crises and the resilience of the overall financial system. The first set of stress tests as mandated by the DFA were required for banks with assets of at least \$50 billion and had to be completed by September 30, 2012. However, the Final Rule of the DFA required stress tests for all banks with assets of at least \$10 billion beginning in the following year (Federal Register 2012). Summary results of the stress tests are publicly disclosed and are closely watched by market participants.

The Dodd-Frank Act stress tests (hereafter DFAST) are designed to gauge bank capital adequacy following potential economic downturns and to assess bank risk taking. The Federal Reserve determines the scenarios to reflect the possible paths of aggregate economic variables, given current economic conditions. Moreover, the Federal Reserve develops a model to analyze each bank's capital adequacy and loan portfolio risk in light of the hypothetical situations. Both the Federal Reserve's model and their hypothetical scenarios for the stress tests are revised annually to reflect changes in economic conditions, as well as in response to richer data or other modeling enhancements that the Federal Reserve sees fit.⁸ As such, banks cannot easily predict their outcomes to the stress tests, and the results are likely informative for banks regarding their ability to withstand plausible declines in economic conditions. The results of the stress tests include projections of capital ratios, loan losses across several loan types, risk-weighted assets, revenues, losses, and net income.

Following the stress tests, projected declines in capital or increases in loan losses (and the corresponding pressure from regulators) can incentivize banks to engage in risk management measures. (We exploit heterogeneity in stress test outcomes in some of our analyses.) These include reducing their current loan portfolio risks or improving their capital adequacy ratios to ensure that they have enough capital on hand in case of adverse economic events. To this end, banks can lower the amount of credit provided or demand higher rates from riskier borrowers. Consistent with this argument, Acharya, Berger, and Roman (2018) and Cortés et al. (2020) document that credit supply was negatively impacted among stress-tested banks. In particular, stress-tested banks significantly increased loan spreads (defined as the interest rate over LIBOR) and reduced lending to risky borrowers, and also maintained higher capital ratios in response to the stress tests.

⁷The Federal Reserve uses current economic conditions to determine potential negative trajectories for the US economy. For example, with respect to the 2020 stress tests, the Federal Reserve announced: "The DFAST 2020 supervisory scenarios include trajectories for 28 variables. These include 16 variables that capture economic activity, asset prices, and interest rates in the US economy and financial markets, and 12 variables made up of 3 variables (real gross domestic product (GDP) growth, inflation, and the US/foreign currency exchange rate) for each of 4 countries/country blocks. [...] The severely adverse scenario is characterized by a severe global recession accompanied by a period of heightened stress in commercial real estate and corporate debt markets" (Board Gov. Fed. Reserve Syst. 2020). For more details, see <https://www.federalreserve.gov/publications/june-2020-supervisory-scenarios.htm>.

⁸As noted in the recent 2021 DFAST report by the Federal Reserve, "Each year, the Federal Reserve refines both the substance and process of the supervisory stress test, including its development and enhancement of independent supervisory models" (p. 19). As an example to changes in the hypothetical scenarios, in 2021 the Federal Reserve made several adjustments due to the COVID-19 event, which include revisions on the default probability or losses associated with auto and credit card loans, commercial real estate, and first-lien mortgages (Board Gov. Fed. Reserve Syst. 2021).

We note that the Federal Reserve implemented other policies related to stress tests around this time. We discuss these other programs further and examine their potential effects in online Appendix A.2.

Hospital Borrowing

Hospitals of all types rely partially on debt to finance their operations. Indeed, borrowing represents one of the sole avenues of external financing for hospitals, as few hospitals have access to public equity markets. Accordingly, almost all of the hospitals in our sample (93 percent) have debt financing, and the amount of debt that hospitals utilize is substantial.⁹ Moreover, hospitals are particularly risky borrowers. For example, health care bonds have significantly higher yields and lower ratings than non-health care bonds (Gao, Lee, and Murphy 2022). Furthermore, health care bonds accounted for 20 percent of all municipal bond defaults from 1999 to 2010 (Gao, Lee, and Murphy 2019).¹⁰ Bank loan maturity periods are also substantially lower for hospitals relative to other industries, which is consistent with evidence that banks tend to provide shorter-maturity loans to riskier borrowers (e.g., Strahan 1999). Therefore, in line with the evidence that banks tend to reduce credit supply to risky borrowers following heightened risk-management incentives induced by stress tests (Acharya, Berger, and Roman 2018; Cortés et al. 2020), banks may be inclined to reduce credit to risky hospital borrowers or to raise interest rates following stress tests.

Due to their reliance on debt as a source of operational financing, hospitals regularly take out new loans even prior to the maturity of existing debt. Thus, credit may be curtailed or more expensive from a hospital's existing lender. Hospitals may alternatively react to this credit shock by seeking credit from other lenders. However, as has been well established in the banking literature, long-term lending relationships help to lower asymmetric information between borrowers and lenders, thus reducing the cost of credit for borrowers.¹¹ New lenders without an established relationship would thus require higher interest rates or provide less credit as a result of greater information asymmetry. Indeed, in line with this argument, we show that after a bank is stress-tested, the hospitals that borrowed from it experience a significant increase in loan spreads, a decrease in loan amounts, and are more likely to borrow from a new lender.¹² These results reinforce the findings of Acharya, Berger, and Roman (2018) and Cortés et al. (2020) and are consistent with the argument

⁹Debt comprises 33 percent of total hospital assets and 58 percent of total liabilities for the average hospital with debt in our sample.

¹⁰Nonprofit hospitals may borrow through tax-free municipal bonds to finance construction for specific infrastructure projects, however this option is not available for most for-profit hospitals. Health care municipal bonds have an average yield of 3.22 percent, while the average for non-health care municipal bonds is 2.39 percent (Gao, Lee, and Murphy 2022).

¹¹For example, see Rajan (1992); Petersen and Rajan (1994); Boot and Thakor (2000); Degryse and Ongena (2005); Bharath et al. (2007); and Botsch and Vanasco (2019); among many others. Boot (2000) and Elyasiani and Goldberg (2004) provide surveys.

¹²In untabulated tests, we also show that the increased cost of credit that hospitals face is not mitigated over time for affected hospitals that take out new loans in the years after their bank is stress-tested. This suggests that the stress test-induced risk management incentives that push banks to restrict credit supply to hospitals is not a short-lived phenomenon.

that hospital borrowers experienced a shock to credit supply subsequent to a lender's stress test.

Consequently, following a shock to credit supply, hospitals may be faced with less external financing or a higher cost of debt. Hospitals operate on thin profit margins and thus the tighter credit constraints are likely to strain hospital finances. As a result, hospitals can respond with revenue-increasing or cost-saving measures.¹³ Since patients are the primary source of revenue, hospitals may be inclined to increase revenues through higher resource utilization, such as increased inpatient admissions or more intensive use of outpatient hospital services.¹⁴ Anecdotal evidence of such actions to increase hospital revenues has been noted in recent media coverage. For example, executives of a major Philadelphia hospital attempted to increase revenue “based in part on the assumption that increasing in-patient admissions through the ER would yield greater reimbursements from insurance companies” (*The New Yorker* June 7, 2021). Moreover, the US Department of Justice has successfully pursued cases involving unnecessary or excessive hospital admissions, including a case where hospitals implemented inpatient admission targets from the hospital emergency department (ED) and even tied admission percentages to physician compensation as a way to increase revenues through greater admissions.¹⁵ Indeed, hospitals have considerable discretion on whether to admit patients who arrive at the ED or have them discharged without an inpatient stay.

Hospitals can also increase admissions by strengthening ties with physician offices, thereby increasing physician referrals for inpatient or outpatient elective

¹³ It is possible that hospitals could simply postpone financing given the higher cost of credit. However, a hospital that is in need of financing would then have to find an alternative (and potentially more expensive) source for such financing or rely more on their own internally generated funds. We would therefore expect a similar response by hospitals that postponed financing in light of the tighter credit constraints (e.g., increasing revenues to cover the higher cost of external financing or the greater reliance on internal funds). Moreover, we find that new loans initiated by affected hospitals following the shock have higher rates and worse terms and that the majority (79 percent) of affected hospitals take out new loans following the shock, suggesting that hospitals cannot afford to substantially delay access to credit and have few alternatives for external financing.

¹⁴ Medicare payments to hospitals are based on the inpatient/outpatient prospective payment system. Specifically, inpatient revenues are determined by the diagnosis-related group (DRG) that the patient is assigned to when admitted, with riskier or more complicated groups corresponding to higher payment rates. Hence, increased admissions or assigning patients to higher-paying DRGs can generate higher revenues. Similarly, outpatient service payments are set prospectively based on ambulatory payment groups. A new procedure gets paid for the ambulatory group it is assigned. As a result, unlike inpatient services, additional procedures for outpatient services can generate more revenue for the hospital.

¹⁵ See, for example, “Hospital Chain Will Pay Over \$260 Million to Resolve False Billing and Kickback Allegations; One Subsidiary Agrees to Plead Guilty,” US Department of Justice Office of Public Affairs, September 25, 2018. As noted by the Department of Justice: “According to admissions made in the resolution documents, HMA instituted a formal and aggressive plan to improperly increase overall emergency department inpatient admissions at all HMA hospitals, including at Carlisle Regional Medical Center. As part of the plan, HMA set mandatory company-wide admission rate benchmarks for patients presenting to HMA hospital emergency departments—a range of 15 to 20 percent for all patients presenting to the emergency department, depending on the HMA hospital, and 50 percent for patients 65 and older (i.e., Medicare beneficiaries)—solely to increase HMA revenue. HMA executives and HMA hospital administrators executed the scheme by pressuring, coercing and inducing physicians and medical directors to meet the mandatory admission rate benchmarks and admit patients who did not need inpatient admission through a variety of means, including by threatening to fire physicians and medical directors if they did not increase the number of patients admitted” (US Department of Justice 2018). We note that Health Management Associates (HMA) is not a treated hospital system in our sample and that the practices described took place prior to our treatment period. Our results are also unaffected if we exclude HMA from the sample.

procedures. Indeed, numerous hospitals have been penalized for providing remuneration to physician offices for patient referrals.^{16,17}

Other revenue-increasing actions include properly documenting the severity of patient conditions to increase reimbursement from insurers, such as hiring nurse-consultants to oversee diagnoses or employing doctors with greater familiarity in medical coding (as noted in Pomorski 2021). Such actions may be considered “upcoding” if patient conditions are exaggerated in insurance claims (Silverman and Skinner 2004; Geruso and Layton 2020). Furthermore, recent anecdotal evidence indicates that hospitals can turn to converting pediatric beds to adult units to increase profitability, as adult admissions typically generate higher reimbursement rates than pediatric admissions. As noted in recent media coverage: “Hospitals around the country, from regional medical centers to smaller local facilities are closing down pediatric units. The reason is stark economics: Institutions make more money from adult patients” (Baumgartner 2022).

Hospitals can also take cost-saving measures, such as delaying new equipment purchases and capital investment, more aggressively pursuing unpaid invoices, or reducing hospital staff.¹⁸ Likewise, for inpatient admissions with reimbursements based on a prospective, fixed payment structure, hospitals can improve cost efficiency by scaling back on services provided during inpatient stays or reducing the length of inpatient stays. Such hospital responses do not suggest a clear prediction on actual patient health outcomes. In particular, greater admitted volume may lead to less attention and thus worse quality of care (e.g., Silver 2021). On the other hand, if inpatient and outpatient services of elective, nonemergency procedures (such as hip and knee replacements) are increased to compensate for the decreased funds, then patient health may be unaffected (Clemens and Gottlieb 2014; Einav, Finkelstein, and Mahoney 2018) or even improved if such measures imply greater attention and care.

II. Research Design

A. Data and Summary Statistics

We utilize data on hospital characteristics and outcomes from a variety of sources.¹⁹ Medicare-certified hospitals (providers), which include almost all hospitals in the United States, are required to submit an annual cost report to a Medicare administrative contractor, in which they provide complete information on facility characteristics. The US Centers for Medicare and Medicaid Services (CMS), a federal agency within the Department of Health and Human Services, maintains the cost report data in the Health Care Provider Cost Reporting Information System

¹⁶Examples include “Hospital Chain Will Pay over \$513 Million for Defrauding the United States and Making Illegal Payments in Exchange for Patient Referrals; Two Subsidiaries Agree to Plead Guilty” (US Department of Justice 2016), and “West Virginia Hospital Agrees To Pay \$50 Million To Settle Allegations Concerning Improper Compensation To Referring Physicians” (US Department of Justice 2020).

¹⁷Hospitals can also directly acquire physician practices or community hospitals to increase referrals (Nakamura, Capps, and Dranove 2007).

¹⁸For anecdotal evidence of hospitals increasing collections from unpaid patient invoices to boost profits, see “They Were Entitled to Free Care. Hospitals Hounded Them to Pay,” *New York Times*, September 24, 2022.

¹⁹Our hospital data is obtained through the National Bureau of Economic Research (NBER) and Adam Sacarny’s website (Sacarny n.d.a; Sacarny n.d.b).

(HCRIS). We obtain all available reported information on hospitals from the HCRIS database (CMS 2010–2016c). For each provider, this covers common financial statement items such as total assets (*TA*), total liabilities, and net income, among others. In addition, the data include hospital utilization information, including total inpatient discharges, total occupied bed days, and total available bed days.²⁰

Our sample includes yearly hospital observations from 2010 to 2016. Our sample begins in 2010 because it is from this date that our key variables are consistently defined; prior to this, a number of our key variables are missing or defined in an inconsistent way in data reporting.²¹ Financial information is complete in the database for most hospitals up to calendar year 2016. We restrict the sample to include only short-term acute care hospitals (the most common type of hospitals), though our results are robust to including other types of hospitals, as well as controlling for hospital-type fixed effects. We further exclude government-owned hospitals (such as Veterans Affairs hospitals and clinics), as they typically directly receive government financial assistance and have different incentive structures than other hospitals (Duggan 2000). Our final sample includes 3,658 unique hospitals.

To measure hospital care quality, we merge the above information with two other datasets from CMS that provide measures of health outcomes and quality of care. The first measure is the risk-adjusted rate of unplanned readmissions, obtained from the CMS Hospital Compare program (CMS 2010–2016b; Sacarny n.d.a). A readmission is defined as an admission to an acute care hospital within 30 days of discharge from a previous hospital stay. Readmission rates are informative about the efficacy of treatment upon hospitalization and are widely-used measures for quality of care by both government agencies and researchers (Chandra et al. 2016; Beaulieu et al. 2020). A relatively high readmission rate, for example, may imply that the hospital is more likely to have provided inadequate care or misdiagnoses during inpatient stays, resulting in more patients unexpectedly requiring rehospitalization. Readmission rates are provided for all diseases combined and are also separately documented for three key acute conditions: acute myocardial infarction (i.e., AMI or heart attack), heart failure, and pneumonia. We additionally collect risk-standardized 30-day mortality data for patients treated for these conditions, also provided by CMS.²²

We also utilize patient evaluations to measure quality of care from the patient's perspective. In particular, we use the Hospital Consumer Assessment of Health Care Providers and Systems (HCAHPS) data (CMS 2010–2016b; Sacarny n.d.a), which is a patient satisfaction survey required by CMS and is administered to a random sample of adult patients across various medical conditions between 48 hours and six weeks after discharge. The core questions cover the critical aspects of patients' hospital experiences, such as the overall rating of the hospital, efficacy of pain control, whether they would recommend the hospital, communication with nurses and doctors, the cleanliness and quietness of the hospital environment, and discharge

²⁰ An occupied bed day is a day during which a person is confined to a bed and in which the patient stays overnight in a hospital. An available bed day is a day in which a bed is in the facility and can possibly be occupied. This includes all types of beds (general and special care).

²¹ One major change is due to the American Recovery and Reinvestment Act of 2009, which motivated hospitals to adopt a health care information technology (HIT) system. After 2010, total assets include accumulated HIT investment net of depreciation.

²² We provide details regarding the risk adjustment performed by CMS in Section A.3 of the online Appendix.

information. Because rating scales differ across categories, we calculate the proportion of patients that give the highest rating instead of using average scores.²³ All variable definitions are also included in online Appendix Table A.1.

To further explore mechanisms, we supplement the above hospital data with two additional datasets. The first is the State Inpatient Databases (SID) developed for the Health Care Cost and Utilization Project (HCUP) (AHRQ 2010). We identify 15 states that provide data both before and after the stress tests.²⁴ Each unit of observation in the SID is an inpatient encounter that records various treatment and demographic information. We discuss the variables in more detail and how we aggregate these in Section IV. The second supplemental dataset is the process of care scores from the CMS Hospital Compare program (CMS 2010–2016b; Sacarny n.d.a). CMS requires hospitals to submit information on timely and effective treatment which have been linked to improve patient outcomes for certain medical conditions. We examine six measures related to our conditions of focus (acute myocardial infarction, heart failure, and pneumonia) from 2010 to 2014.²⁵ We provide more detail on the measures in Section IV.

Lastly, we combine our hospital data with data on stress-tested banks from the Board of Governors of the Federal Reserve System (2022) and Dealscan loan data (Refinitiv n.d.) in order to identify treated and control hospitals. We keep all loan agreements (facilities) which have (i) a borrower three-digit SIC code equal to 806 (Hospitals); (ii) a facility start date after January 1, 2007; and (iii) loan types that are either term loans or revolver. Following Ivashina (2009), we identify and keep the lead bank in a syndicate deal.²⁶ This results in 2,432 facility-lender combinations. The hospital-related borrowers in Dealscan are either individual providers (e.g., Houston Methodist Hospital) or hospital organizations and systems (e.g., HCA Health Care). We then manually match borrowers to the HCRIS sample using hospital affiliation data from IQVIA (n.d.). For each individual hospital, HCRIS reports whether it belongs to a hospital chain and the organization name if it does. When we identify a borrower that is a hospital system, we assign each of the individual hospitals that are part of the system as being exposed to the loan deal. There are 1,447 facility-lender combinations in which we identify that the borrower is a Medicare-certified hospital.²⁷

²³For example, the survey question for the variable *Recovery Information* is whether the patient was given information about what to do during their recovery at home, where the answer choices are “Yes” or “No.” The question for the variable *Overall Rating* is a star-rating system from 1 (worst) to 3 (best). We define “highest rating” as answering “Yes” in the former and “3” in the latter.

²⁴We access the SID through the National Bureau of Economic Research and utilize hospital identification data from the American Hospital Association (AHA 2010–2016) to match hospitals between datasets. The 15 states with coverage during the sample period through our access include Arizona, Colorado, Florida, Kentucky, Maryland, Massachusetts, Nevada, New Jersey, New York, North Carolina, Rhode Island, Utah, Vermont, Washington, and Wisconsin.

²⁵While the CMS data has other measures, we focus on these six measures because they are the most continuously-tracked and non-missing over our sample period. In 2005, the first set of ten “core” process of care measures were created for acute heart infarction, heart failure, pneumonia, and surgical care. Over the years, the program has terminated existing measures and medical conditions and has added new measures. This makes the other measures infeasible to use for our purposes.

²⁶In our sample, this includes the Dealscan lender roles: admin agent, arranger, documentation agent, senior managing agent, or syndications agent.

²⁷The major borrowers that we do not match include psychiatric hospitals, specialty hospitals, non-Medicare hospitals, and telehealth service platforms.

TABLE 1—LENDER AND EXPOSED HOSPITAL DISTRIBUTIONS AND SUMMARY STATISTICS

	Tested lenders	Existing loans	Exposed hospitals		
	(1)	(2)	(3)		
<i>Panel A. Tested lenders distribution</i>					
2012	15	52	400		
2013	4	26	41		
2014	3	3	27		
2015	1	4	33		
2016	3	4	4		
Total	26	89	505		
	Mean	SD	P25	Median	p75
	(1)	(2)	(3)	(4)	(5)
<i>Panel B. Loan characteristics</i>					
<i>Spread&Fee</i>	388.479	305.089	200.000	325.000	475.000
<i>Maturity</i>	57.603	16.346	49.000	60.000	60.000
<i>Loan Amount</i> (\$ millions)	78.402	312.565	7.955	13.231	25.132
<i>Total Yearly Amount</i>	144.296	271.408	24.168	50.000	133.333

Notes: Panel A summarizes the yearly distribution of first-time stress-tested banks and exposed hospitals. In panel A, column 1 shows the number of new banks that were stress-tested *and* were lending to hospitals in the sample in a given year. Column 2 shows the number of existing loans to hospitals by these newly-tested lenders in each year. Column 3 shows the affected number of hospitals that borrow from the lenders in column 1 in each year. Panel B provides summary statistics for the main loan variables in our sample. *Spread & Fee* is the interest rate spread over LIBOR plus fees on the drawn portion of the loan (in basis points). *Maturity* is the loan facility maturity (in months). *Loan Amount* is the individual facility amount for each hospital borrower (in \$ millions). If the borrower is a hospital chain/system, we divide the facility amount by the number of subsidiary hospitals in the system at the borrowing time. *Total Yearly Amount* is the total amount of new debt (in \$ millions) across individual facilities taken out by a given hospital in a given year.

Panel A of Table 1 shows the yearly number of first-time stress-tested banks along with the exposed hospital borrowers in our sample. From 2012 to 2016, 26 stress-tested banks were lending to at least one hospital in our sample when tested for the first time. In total, this leads to 505 hospitals (out of 3,658) being exposed to the Dodd-Frank Act stress tests (DFAST). Banks with consolidated assets of \$50 billion or above were required to conduct their first annual stress tests using financial data as of September 30, 2012. Given their size, these banks jointly held a significant market share for hospital lending. In our sample, 15 banks (58 percent of the stress-tested banks) and 416 hospitals (82 percent of the affected hospitals) are exposed to the first DFAST that occurred in 2012. Banks with total consolidated assets of more than \$10 billion but less than \$50 billion were required to implement stress tests under the Dodd-Frank Act in the following years. The geographical distribution of affected hospitals is illustrated in Figure 3 (see Section III).

Panel B of Table 1 provides summary statistics for the loan characteristics in our sample. Bank loans are an important source of external financing for hospitals. The size of a bank loan for an individual lending agreement averages to \$78.4 million. Hospitals typically have multiple bank loans outstanding contemporaneously; the average amount of new bank lending taken out in a given year is \$144.3 million for hospitals that utilize bank lending in our sample. Loan maturity periods average just under five years, with an average annual interest rate of 3.88 percent over LIBOR. This amounts to an average of \$7.0 million in interest expenses per year for a given

TABLE 2—SUMMARY STATISTICS FOR FULL SAMPLE AND BY TREATMENT STATUS

	Obs.	Mean	SD	p25	Median	p75
<i>Panel A. Full sample</i>						
<i>Net Income/TA</i>	3,511	0.058	0.226	−0.007	0.040	0.101
<i>log(Income)</i>	3,511	18.237	1.330	17.272	18.317	19.254
<i>Liabilities/TA</i>	3,510	0.584	0.533	0.255	0.483	0.747
<i>Cash/TA</i>	3,461	0.073	0.119	0.002	0.035	0.098
<i>log(Discharges)</i>	3,572	8.091	1.568	6.989	8.378	9.330
<i>log(Bed Days)</i>	3,575	10.276	1.068	9.119	10.411	11.123
<i>For-profit</i>	3,633	0.217	0.412	0.000	0.000	0.000
<i>Hospital System</i>	3,658	0.678	0.467	0.000	1.000	1.000
<i>Panel B. Treatment group</i>						
<i>Net Income/TA</i>	494	0.104	0.234	−0.005	0.065	0.171
<i>log(Income)</i>	494	18.694	1.014	18.104	18.789	19.422
<i>Liabilities/TA</i>	496	0.577	0.630	0.117	0.424	0.813
<i>Cash/TA</i>	485	0.034	0.083	0.000	0.001	0.040
<i>log(Discharges)</i>	496	8.747	1.192	8.267	8.968	9.543
<i>log(Bed Days)</i>	497	10.744	0.855	10.343	10.884	11.338
<i>For-profit</i>	504	0.492	0.500	0.000	0.000	1.000
<i>Hospital System</i>	505	0.988	0.108	1.000	1.000	1.000
<i>Panel C. Control group</i>						
<i>Net Income/TA</i>	3,017	0.051	0.223	−0.007	0.037	0.093
<i>log(Income)</i>	3,017	18.162	1.361	17.142	18.202	19.186
<i>Liabilities/TA</i>	3,014	0.585	0.515	0.272	0.491	0.742
<i>Cash/TA</i>	2,976	0.079	0.123	0.006	0.042	0.105
<i>log(Discharges)</i>	3,076	7.985	1.595	6.858	8.204	9.251
<i>log(Bed Days)</i>	3,078	10.200	1.080	9.119	10.282	11.059
<i>For-profit</i>	3,129	0.172	0.378	0.000	0.000	0.000
<i>Hospital System</i>	3,153	0.628	0.483	0.000	1.000	1.000

Notes: This table provides the summary statistics of hospital characteristics in 2011 (the last year before the DFAST announcement) for the entire sample and separately for the treatment and control groups. All variables are defined at the hospital-year level. *Net Income* is *Income* minus *Cost*. *Cost* is total costs (operating and other expenses). *Income* consists of net patient revenue and other income. *Liabilities* is total liabilities. *Cash* is cash holdings. *TA* is total assets; all variables scaled by *TA* are winsorized at the 1 percent level. *Discharges* is the number of inpatient discharges. *Bed Days* is the number of available beds multiplied by 365. *For-profit* is an indicator variable equal to one if the hospital is a for-profit hospital, and zero otherwise. *Hospital System* is an indicator variable equal to one if the hospital is a part of a hospital system, and zero otherwise.

hospital in our sample that utilizes bank lending, indicative of the high cost of servicing debt for hospitals.

Summary statistics for all of our other variables are provided in online Appendix Table A.1. We also provide summary statistics for our full sample as well as separately for our treatment and control groups in the year before DFAST implementation in Table 2. We additionally show that all of our main results hold for a sample that is tightly matched on observable characteristics; summary statistics for this matched sample are provided in online Appendix Table A.2.²⁸

²⁸Specifically, we construct our treatment and control groups by matching based on the year 2011 values of cash over total assets, logarithm of bed days, patient revenues over total assets, and pneumonia mortality rate based on the nearest two neighbors for each treatment hospital. We restrict our matched sample to a precision difference cutoff of 0.003. Summary statistics for the variable differences between the treatment and control samples in 2011 are provided in Table A.2, and show that our constructed treatment groups are tightly matched in terms of observable characteristics—there is no statistically significant difference between the treated hospitals (hospitals that have lending relationships with stress-tested banks) and control hospitals (those without loans from stress-tested banks).

B. Empirical Specification

For our main specification, we examine a staggered difference-in-differences (DID) regression to explore the effect of bank stress tests on hospital outcomes:

$$(1) \quad Y_{i,t} = \alpha + \beta STExposed_{i,t-1} + \gamma' \mathbf{Controls}_{i,t-1} + \eta_t + \mu_i + \varepsilon_{i,t}.$$

In equation (1), $STExposed_{i,t-1}$ is an indicator variable that takes a value of one if at least one of hospital i 's relationship banks experienced a stress test in year $t - 1$ or earlier, and zero otherwise. Hospital i 's relationship bank is defined as a lending bank that has non-matured loans with hospital i in year t . $Y_{i,t}$ is the outcome variable, which includes measures of hospital financial, operational, and care quality information. The parameters η_t and μ_i denote year and hospital fixed effects, respectively.

In equation (1), we include a vector of lagged control variables, $\mathbf{Controls}_{i,t-1}$, in order to account for hospital-level characteristics that have the potential to drive differences between hospitals with respect to financing and operating decisions. For example, larger hospitals may experience relatively smaller percentage changes year-to-year in profitability or resources, and thus not controlling for size may induce omitted variable bias when examining responses to the credit supply shock. As such, we include $\log(BedDays_{i,t-1})$, which is the logarithm of one plus available bed days,²⁹ to control for size based on physical hospital capacity. Likewise, cash holdings scaled by total assets ($Cash/TA_{i,t-1}$) controls for the ability of a hospital to utilize internal financing (accounting for its size), as a large literature in corporate finance has highlighted that this can influence investment and operating decisions by firms (Opler et al. 1999; Bates, Kahle, and Stulz 2009). We lag all control variables in order to avoid using information not known at the time of the operating decision we examine as the dependent variable.³⁰

The coefficient of interest in equation (1) is β , which captures the effect of bank stress-tests on hospital outcomes. Put differently, β represents the change in hospital outcomes after a stress test exposure in a year relative to the corresponding change for hospital-year observations with no stress test exposure. Our variation in treatment comes from (i) whether the hospital relies on loan financing from a bank that was subject to the DFAST requirements and (ii) the staggered implementation of stress tests for different banks.³¹ A recent econometrics literature has noted issues related to interpreting staggered DID specifications with two-way fixed effects; we

²⁹ A bed means an adult bed or other beds maintained in a patient care area for lodging patients in the hospital, including pediatric beds. Bed days are computed by the number of available beds multiplied by the number of days in the reporting period.

³⁰ The results are qualitatively similar if we exclude control variables from the analysis, as shown in online Appendix Table B.1. We also present parallel trend graphs estimated without control variables in online Appendix Figure B.1. Our results are also very similar when including additional control variables (available upon request). As is standard in the corporate finance literature, we winsorize all financial variables scaled by total assets at the 1 percent level to account for extreme outliers.

³¹ As previously noted, the majority of treated hospitals (82 percent) are first exposed to DFAST in 2012, and thus the former source of variation is more predominant. Our results are robust to specifying our DID tests through a non-staggered specification by defining a dummy variable that takes a value of one if the year is 2013 or later (and zero otherwise), and interacting this with a treatment variable at the hospital level that takes a value of one if a hospital had a relationship loan prior to DFAST implementation (and zero otherwise).

discuss this in Section IIID and show that our results are robust to methods which explicitly account for these issues.

The identifying assumption is that a stress test to an affected bank is exogenous to the performance of the hospital which has a relationship with that specific bank. Reverse causality is not likely to hold in this setting, since the DFAST did not select a participating bank based on the hospitals which borrowed from the bank. In particular, banks were selected to be stress-tested based on whether their total assets exceeded a \$10 billion threshold, which is exogenous to the hospitals which borrowed from the banks.

Self-selection by hospitals is also not likely to happen. Although the Dodd-Frank Act was enacted on July 21, 2010, the FDIC issued a notice of proposed rule-making (NPR) on January 23, 2012. This NPR solicited public comment to finalize the implementation of the Act, and the effective date and public disclosure policy of results were changed due to major concerns. Thus, the actual timing of DFAST implementation was uncertain and therefore exogenous to the loan initiation. A hospital also had no incentive to borrow from a particular bank based on the fact that this bank would be stress-tested soon; indeed, as the relationship lending literature has shown, relationship borrowers tend to choose lenders based on factors such as whether the bank operates in the same geographical area (e.g., Petersen and Rajan 1995; Cantillo and Wright 2000; Degryse and Ongena 2005).³² Most stress-tested banks operate nationally with branches located across different states. Our discussions with senior hospital executives confirm this notion and provide anecdotal evidence that bank branch proximity is a major factor for hospitals when determining lending relationships. In addition, the vast majority of hospital-bank lending relationships in our sample were established prior to DFAST.³³

We further validate our inferences related to these arguments in a number of ways. First, we show that the parallel trends assumption holds in our setting. Second, as mentioned previously, for all of our main tests, we demonstrate that our main results hold when restricted to a sample of hospitals tightly matched in terms of observables. We also show that the parallel trends assumption holds for this matched subsample. Finally, we include a host of robustness checks related to controlling for potential differences between hospitals as well as other sample composition effects.

III. Results

A. Stress Tests and Credit Supply

We begin our analysis by examining the effect of stress tests on hospital loans. While Acharya, Berger, and Roman (2018) and Cortés et al. (2020) have previously shown that stress tests negatively impact credit supply, we investigate whether these

³² We note that affected hospitals do not concentrate in certain areas, as shown by Figure 3.

³³ We exploit heterogeneity in bank loan financing reliance in further tests. Moreover, in untabulated tests, we confirm that the parallel trends assumption holds for the subsample of hospitals that rely more on bank financing. This reinforces the notion that it is specifically variation related to *bank* loans and shocks to specific lenders that are affecting treated hospitals. We also show that the effects hold when restricting our sample to only hospitals that received commercial loans during the sample period (online Appendix Table A.22).

effects are present for our sample of hospital borrowers as well. To do so, we estimate equation (2) at the *loan facility* level:

$$(2) \quad Y_{k,i,j,t} = \alpha + \beta STExposed_{i,t-1} + \gamma' \mathbf{Controls}_{i,t-1} + \mu_j + \eta_t \\ + TypeFE + PurposeFE + \varepsilon_{k,i,j,t}.$$

The variable $Y_{k,i,j,t}$ represents the characteristics of loan k between hospital i and bank j which was originated in year t . These characteristics include the loan spread and fee, amount, and maturity. $STExposed_{i,t-1}$ is equal to one if hospital i borrowed from a bank that was stress-tested in year $t - 1$ or earlier, and thus has been indirectly exposed to the stress tests. We note that outcome Y is measured for each loan k between hospital i and bank j , but the value of $STExposed_{i,t}$ is determined by hospital i 's exposure and is independent of the particular lender j in this loan. For example, consider a hospital i that has a lending relationship with a stress-tested bank j' in year $t - 1$. If hospital i switches to a *new* lender j (potentially untested) in year t , then $STExposed_{i,t} = 1$ for this deal between i and j . This specification therefore allows us to capture the possibility that the hospital switches to a new bank with potentially different loan characteristics (e.g., higher spread). As noted in Section I, starting a relationship with a new lender generally entails a higher cost of debt to compensate for the greater degree of asymmetric information. We include control variables for the hospital's logarithm of total assets, patient revenues over total assets, leverage (total liabilities over total assets), and tangibility (total fixed assets over total assets). We also include bank (μ_j), year (η_t), loan type (*TypeFE*), and loan purpose (*PurposeFE*) fixed effects. Following Drucker and Puri (2009), loan types include *Revolvers* and *Term Loans*. Loan purposes include *Acquisition*, *General*, *LBO*, *Recapitalization*, *Miscellaneous*, and *Other*.

Table 3 provides the results. In columns 1 and 2, we examine loan interest rates, defined as the spread (in basis points) over LIBOR plus one-time fees on the drawn portion of the loan. We see that borrowing costs (i.e., interest rates on the loans) for affected hospitals increase significantly by 68–74 basis points, an increase of about 17.5–19 percent of the sample average of 3.88 percent (Table 1). In columns 3 and 4, we find that loan size decreases by 36.5 percent and loan maturity decreases by 8.5 percent for affected hospitals. These results are consistent with Acharya, Berger, and Roman (2018) and Cortés et al. (2020) and suggest that hospital credit access was negatively impacted by stress testing, as exemplified through a higher cost of debt and lower loan amounts and maturities for affected hospitals. Moreover, the large impact of the credit shock on loan interest rates and amounts underscores the riskiness of hospital borrowers (as noted previously, hospitals exhibit higher default rates compared to other industries).

In column 5, we consider the possibility that hospitals may switch lenders following a stress test, as switching to a new lender with which the borrower has no relationship entails a higher cost of credit (Ostromogolsky 2016). To explore this, we define the variable *New Lender* $_{k,i,j,t}$, which is an indicator variable equal to one if hospital i had no previous lending relationship with bank j prior to year t . The coefficient on *New Lender* $_{k,i,j,t}$ is positive and significant, which implies that

TABLE 3—HOSPITAL LOAN CHARACTERISTICS

	<i>Spread&Fee</i> (1)	<i>Spread&Fee</i> (2)	$\log(\text{Loan Amount})$ (3)	$\log(\text{Maturity})$ (4)	<i>New Lender</i> (5)
<i>STExposed</i> _{<i>i,t-1</i>}	74.023 (25.206)	68.235 (29.585)	-0.365 (0.128)	-0.085 (0.049)	0.126 (0.073)
<i>Controls</i>	N	Y	Y	Y	Y
<i>Year</i> fixed effects	Y	Y	Y	Y	Y
<i>Bank</i> fixed effects	Y	Y	Y	Y	Y
<i>Loan type</i> fixed effects	N	Y	Y	Y	Y
<i>Loan purpose</i> fixed effects	N	Y	Y	Y	Y
<i>N</i>	1,052	717	810	801	810
<i>Y</i> mean	388.479	388.479	78.402	57.603	0.420
<i>Y</i> SD	305.089	305.089	312.565	16.346	0.494
Adj <i>R</i> ²	0.15	0.31	0.60	0.44	0.34

Notes: This table provides the regression results for equation (2). Each observation represents a loan facility *k*, borrowed by hospital *i* from bank *j* in year *t*. *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* - 1 or earlier, and 0 otherwise. *Spread & Fee* is the interest rate (in basis points) spread over LIBOR plus fees on the drawn portion of the loan. $\log(\text{Loan Amount})$ is the logarithm of the loan facility amount. $\log(\text{Maturity})$ is the logarithm of the loan facility maturity (in months). *New Lender* takes a value of 1 if hospital *i* has never borrowed from bank *j* before year *t*, and 0 otherwise. Control variables include borrower *i*'s logarithm of total assets, income over total assets, liabilities (total liabilities over total assets), and tangibility (total fixed assets over total assets). Year, bank, loan type, and loan purpose fixed effects, are included as indicated. The mean and standard deviation for each dependent variable (denoted *Y*) are reported (presented nonlogged if the dependent variable is a logarithm). Standard errors are clustered at the lender level and provided in parentheses.

hospitals are 12.6 percent more likely to switch to new lenders when their current lender is subject to a stress test.

The above results evaluate the impact of stress tests in a sample of hospital loans. A related question is how stress-tested banks change their lending terms to hospitals relative to other firms. In Table A.3 of the online Appendix, we add deals by public *non-hospital* borrowers to the initial sample in Table 3.³⁴ We find that, while stress-tested banks increase the spread for all firms by 14 basis points, the spread increases by an additional 35 basis points for hospital borrowers. Moreover, we find a significant decrease in the loan amount only for hospital deals. The more severe response to credit supply for hospital borrowers is consistent with prior evidence that hospitals are particularly risky borrowers. The heightened risk management incentives from banks following the stress tests therefore results in banks imposing a greater cost of debt on hospitals relative to other, less risky borrowers. For robustness, in columns 3 and 4 of Table A.3, we restrict the deals to banks that have previously provided loans to hospitals in our sample. The sample size drops by around only 10 percent, which helps to alleviate the possible selection concern that only a small group of banks specialize in hospital loans.

Finally, we do not find any changes in the health care municipal bond market occurring at the time of the stress tests at the county level for affected hospitals. Specifically, we find that the rates of return, and thus the cost of borrowing, related to health care municipal bonds in a county with affected hospitals do not change,

³⁴ We focus on public firms for comparison in order to include financial and other firm characteristics from the Compustat database (S&P Global Rankings n.d.) as control variables, although we obtain similar conclusions when we include all other firms with loans and exclude firm characteristics as controls.

which suggests that the shock to credit access is specifically related to *bank* borrowing and is not part of a broader shock to hospital borrowing in other debt channels. (We provide these results in online Appendix Table A.4.³⁵) Collectively, these results validate the use of DFAST as a negative shock to hospital credit access.

Overall, these results are consistent with the notion that stress testing negatively impacts credit supply for hospitals. As discussed in Section I, banks subject to stress tests are more inclined to improve their capital adequacy ratios by raising interest rates or lowering loan amounts. Moreover, hospitals may turn to new lenders to make up the loss in credit access, but, due to higher information asymmetries, face higher interest rates in these loans from new lenders.

B. Hospital Financing and Operating Decisions

We now examine the direct impact that the negative credit supply shock had on hospital financing and operating decisions for affected borrowers. We then investigate the indirect impact these decisions had on quality of care and patient health outcomes in the following section.

As discussed in Section IIA, we consider key financial statement information provided through the HCRIS database. The results, which employ specification (1), are presented in Table 4. We first consider the effect of the credit shock on the overall profit margin of the hospital.³⁶ In column 1 of Table 4, panel A, we find that affected borrowers saw a significant increase of 1.1 percentage points in their profit margins. We next examine leverage and other liabilities in column 2. We see a significant reduction in liabilities for affected hospitals. This suggests that affected hospitals utilize debt less following the credit shock, consistent with our previous results in Table 3 indicating lower amounts for new loans. Finally, we consider the level of net income (i.e., profit) scaled by total assets in column 3. The results are consistent with a shift in financing sources. Affected hospitals become both more efficient and generate significantly higher internally generated funds following the tightened credit constraints, bolstering internal financing with a lower reliance on debt financing.³⁷

To provide additional texture to these results, a back of the envelope calculation using column 1 of Table 3 indicates an increase of \$1.07 million in interest payments for the average affected hospital (a sizable increase of 19 percent relative to the sample mean).³⁸ This increase is also economically meaningful for affected

³⁵ Online Appendix Table A.4 explores bond yields, which represents the average rate of return that an investor can expect from purchasing the bond, as well as spreads, which represents the difference between the average bond rates of return and low-risk benchmark investments (US Treasury bonds and highly-rated corporate bonds). Data on municipal bond yields come from the Mergent Inc. (n.d.) Fixed Income Securities Database. All of the coefficient estimates are insignificant and small in magnitude; for example, a county with an exposed hospital experiences an insignificant 0.023 percentage point change in health care municipal bond yields.

³⁶ Profit margin is defined as profit (i.e., net income) divided by *Income*, which is defined as net patient revenues (Worksheet G-3, line 3) plus total other income (line 25). Hospital profit is taken from line 29 of HCRIS Worksheet G-3 and is defined as *Income* minus *Cost*, where *Cost* is defined as operating expenses (line 4) and other expenses (line 28). Profit margin is winsorized at the 1 percent level.

³⁷ We scale net income by total assets to account for size; we cannot take the log of this variable due to the presence of negative values. We additionally show in Table A.5 that the increase in net income is driven by an increase in income generated from patients.

³⁸ The average loan amount per hospital across lending agreements in a given year is \$144.30 million. The increase in interest expense is therefore calculated as $\$144.30 \text{ million} \times 74 \text{ basis points} = \1.07 million .

TABLE 4—HOSPITAL FINANCIAL AND OPERATIONAL PERFORMANCE

	<i>Profit Margin</i>	<i>Liabilities/TA</i>	<i>Net Income/TA</i>	<i>Bed Utilization</i>	<i>Discharge Rate</i>
	(1)	(2)	(3)	(4)	(5)
<i>Panel A. Main specification</i>					
<i>STExposed</i> _{<i>i,t-1</i>}	0.011 (0.006)	-0.082 (0.016)	0.012 (0.007)	0.023 (0.004)	2.398 (0.409)
<i>Controls</i>	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y
<i>N</i>	23,804	23,231	23,233	23,269	23,267
<i>Y mean</i>	0.032	0.565	0.053	0.443	41.933
<i>Y SD</i>	0.283	0.516	0.124	0.231	19.545
<i>Adj R</i> ²	0.21	0.76	0.55	0.94	0.80
<i>Panel B. Propensity score matched sample</i>					
<i>STExposed</i> _{<i>i,t-1</i>}	0.024 (0.007)	-0.081 (0.021)	0.020 (0.009)	0.022 (0.005)	2.196 (0.566)
<i>Controls</i>	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y
<i>N</i>	6,919	6,775	6,770	6,785	6,785
<i>Adj R</i> ²	0.29	0.78	0.57	0.92	0.77

Notes: This table provides the regression results for equation (1) for financial and operational outcome variables. *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* - 1 or earlier, and 0 otherwise. *Profit Margin* is profit margin, defined as $(Income - Cost)/Income$. *Liabilities/TA* is total liabilities over total assets. *NetIncome/TA* is hospital net income $(Income - Cost)$ over total assets. *Bed Utilization* is the average daily fraction of hospital beds that are occupied. *Discharge Rate* is inpatient discharges over total bed days. Control variables include the lagged logarithm of one plus available bed days ($\log(BedDays_{i,t-1})$), and lagged cash holdings over total assets ($Cash/TA_{i,t-1}$). Panel B replicates the analysis using a propensity score matched sample. Year and hospital fixed effects are included, as indicated. The mean and standard deviation for each dependent variable (denoted *Y*) are reported (presented nonlogged if the dependent variable is a logarithm). Standard errors are clustered at the hospital level and provided in parentheses.

hospitals. For example, average hospital profit in our sample is \$8.01 million per year, and thus the increase in interest expense constitutes 13.4 percent of net income (17.4 percent for the median hospital), representing a sizable drop in profits.³⁹ To provide a comparison with this estimate, we consider the raw change in net income and find that the average affected hospital increases its net income by about \$1.12 million following the credit shock.⁴⁰ There are a few potential explanations for why hospitals may boost internally generated funds by more than the higher cost of debt. First, the hospital may recognize the need to rely less on debt in future periods, as shown in column 2 of Table 4. Second, a hospital that expects to have less favorable

³⁹ Net income for the median hospital is \$2.12 million. The median loan amount per hospital across lending agreements in a given year is \$50 million, which implies an interest expense increase of \$0.37 million for the median hospital. As a fraction of median net income, this increase is 17.4 percent.

⁴⁰ For the purpose of magnitude comparison, we calculate this change in net income by running our main specification using raw net income as the dependent variable; we consider both total net income (line 29 of Worksheet G-3), which corresponds to an increase of \$1.12 million, and net income from patient services (line 5), corresponding to an increase of \$1.48 million. These estimates are included in online Appendix Table A.5. We use levels as we cannot take the log of the profit variable due to the presence of negative values, which is not a concern for our other variables. Instead, to reduce the influence of outliers, for our magnitude discussions related to profit we winsorize profit at the 5 percent level (e.g., Dranove, Garthwaite, and Ody 2017).

debt financing terms in future periods can improve income in the *current* period to build sufficient internal reserves for the future.

To further examine the potential sources of the increase in profitability, we consider bed utilization and discharge rates in columns 4 and 5 of Table 4. *Bed Utilization* in column 4 represents the utilized hospital bed days over all available bed days. In other words, it is the fraction of time that a hospital bed is used in a given year. *Discharge Rate* in column 5 is the number of total inpatient discharges in a year over total available beds. Hence, this measure represents the number of patients using each hospital bed in a year. We see that both bed utilization and discharge rates significantly increase for affected hospitals, with these hospitals accommodating 2.4 more patients per bed, equivalent to 191 more patients per hospital per year.⁴¹ Likewise, each hospital bed is occupied 8.4 more days per year, which amounts to 668 more occupied bed days a year per affected hospital.⁴²

For robustness, panel B of Table 4 provides the estimates using a propensity score-matched sample. We see that the results are very similar in this analysis. This suggests that potential differences between the control and treatment groups are not materially affecting the inferences of our analysis. Relatedly, our results are similar in a subsample restricted to only for-profit hospitals (Table 12), providing further evidence that the results are not being driven by differences in ownership type.

We explore additional operating and investing decisions in Table A.5 of the online Appendix. As discussed previously, impediments to credit access can constrain expenditures for capital investment. In line with this notion, we find that the book value for buildings significantly decreases for affected hospitals.⁴³ Furthermore, the negative credit supply shock may constrain hospitals from investing in new equipment. We find that hospital tangible assets (total fixed assets over total assets) insignificantly decreases after exposure to a stress-tested lender. Relatedly, hospitals can accommodate more patients by increasing capacity; however, we do not see a significant increase in the total number of beds within affected hospitals, which suggests that hospitals cannot easily increase bed capacity. Hospitals can also increase profitability by more aggressively pursuing unpaid patient invoices. We find a significant decrease in bad debt expense for affected hospitals, which is consistent with fewer write-offs of expected patient collections.⁴⁴ Finally, we examine the disparate sources of revenues and find an increase in both inpatient and outpatient revenues, with the former exhibiting a larger magnitude, in line with our earlier results of increases in bed utilization and discharge rates.

Put together, these results suggest that hospitals which experience a negative credit supply shock—and thus reduced financial slack—respond by changing their

⁴¹ This is calculated by multiplying 2.4 with the average number of hospital beds in our sample (79.5).

⁴² The coefficient from column 4 indicates an increase in bed utilization of 2.3 percentage points, which translates to $0.023 \times 365 = 8.4$ additional days of bed occupancy in a year. Hospitals in our sample have an average of 79.5 beds, leading to an increase across beds of 668 days.

⁴³ These effects are consistent with previous studies showing a reduction in investment following a negative credit shock (Campello, Graham, and Harvey 2010; Duchin, Ozbas, and Sensoy 2010; Gropp et al. 2019; Dwenger, Fossen, and Simmler 2020).

⁴⁴ We use bad debt expense and not uncompensated care because CMS modified the cost report instructions in 2017, which changed how hospitals calculate their uncompensated care costs. Hospitals were allowed to retroactively adjust their uncompensated care costs for 2015 and 2016, although not all hospitals did. This change in measurement makes the uncompensated care variable unsuitable for time-series analysis. See Medicaid and CHIP Payment and Access Commission (2021) for details.

operations. By accommodating more patients, as evidenced by increased bed utilization and discharges, hospitals are able to increase their revenues generated by patients and profitability on the margin. This is consistent with other papers that have shown an increase in financial efficiency for borrowers following tightened financial constraints (Hovakimian 2011). In Section IV, we utilize additional data to provide further evidence on the underlying mechanisms for the observed increase in profitability and bed utilization. While the operational changes employed by affected hospitals can improve profit margins, these changes may not improve patient care. We explore this question in the next section.

C. Patient Health Outcomes and Quality of Care

We now investigate whether the shock to credit supply indirectly affected patient health outcomes and quality of care. As noted in Section I, increased inpatient admissions and outpatient services and tests may improve quality of care if this implies greater attentiveness. Conversely, the increase in admissions and procedures may be clinically unnecessary and thus may not affect, or could worsen, patient outcomes. For example, a greater volume of patients more severely strains staff and physician time, which can lead to less attention and a lower quality of care.

We consider several measures of health outcomes and care quality to explore our central research question. We first examine the impact on hospital performance using readmission rates. We then investigate whether changes in performance adversely affect patient mortality outcomes. Finally, we consider the potential effect on patient experiences through the patient satisfaction surveys.

Readmission and Mortality.—Our primary measures of health outcomes are unplanned, risk-standardized readmission rates, which track unplanned inpatient readmissions to a hospital within 30 days from discharge from a previous hospital stay. As noted in prior studies, these measures reflect adequacy of care; a patient that was treated properly in the original admission is less likely to be unexpectedly in need of care shortly following discharge. The results are presented in panel A of Table 5 (we also include parallel trend figures, discussed in the following section). Columns 1–3 present the logarithm of the number of patients readmitted within 30 days who were diagnosed with pneumonia, heart failure, or AMI, respectively. We see significant increases across all three measures. The effects are also economically large. Using the point estimates from panel A, affected hospitals have a 10.2 percent increase in unplanned pneumonia readmissions relative to unaffected hospitals and a 3.0 percent increase in both heart failure and AMI readmissions. This translates to an additional 1,570 patients readmitted per year indirectly due to the negative credit shock in aggregate across affected hospitals.⁴⁵

Columns 4–6 consider the rates of unplanned readmissions, which captures the per-patient likelihood of being readmitted for each medical condition, and shows a similar effect: across all three diagnostic groups, we see a 0.3 percentage point

⁴⁵ We calculate this number based on the unconditional means for readmissions of each diagnostic group (reported in Table A.1) and their estimated percentage increases among affected hospitals from Table 5. At the individual hospital level, we observe 3.11 more readmissions per affected hospital per year.

TABLE 5—HOSPITAL CARE QUALITY: READMISSION AND MORTALITY RATES

	$\log(\text{Pneumonia}$ Read- $\text{missions})$ (1)	$\log(\text{Heart}$ Failure Read- $\text{missions})$ (2)	$\log(\text{AMI}$ Read- $\text{missions})$ (3)	Pneumonia Read- mission Rate (4)	Heart Failure Readmission Rate (5)	AMI Read- mission Rate (6)	All Read- mission Rate (7)	All Read- missions Worst (8)
<i>Panel A. Readmission</i>								
$STExposed_{i,t-1}$	0.102 (0.012)	0.030 (0.011)	0.030 (0.013)	0.003 (0.001)	0.003 (0.001)	0.003 (0.001)	0.002 (0.000)	0.047 (0.013)
<i>Controls</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>N</i>	21,611	20,078	12,674	23,432	22,186	14,349	17,694	19,354
<i>Y mean</i>	18.953	26.100	13.062	0.173	0.225	0.174	0.155	0.079
<i>Y SD</i>	16.495	26.645	12.609	0.014	0.019	0.017	0.010	0.270
<i>Adj R²</i>	0.96	0.98	0.96	0.72	0.77	0.82	0.67	0.48
	$\log(\text{Pneumonia}$ Mortality $\text{Num})$ (1)	$\log(\text{Heart}$ Failure Mortality $\text{Num})$ (2)	$\log(\text{AMI}$ Mortality $\text{Num})$ (3)	Pneumonia Mortality Num (4)	Pneumonia Mortality Rate (5)	Pneumonia Mortality Worst (6)		
<i>Panel B. Mortality</i>								
$STExposed_{i,t-1}$	0.098 (0.012)	0.003 (0.010)	0.022 (0.011)	1.737 (0.336)	0.002 (0.001)	0.018 (0.010)		
<i>Controls</i>	Y	Y	Y	Y	Y	Y		
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y		
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y		
<i>N</i>	21,566	19,850	14,073	21,566	23,396	23,844		
<i>Y mean</i>	14.512	11.226	9.398	15.307	0.139	0.053		
<i>Y SD</i>	12.997	10.601	8.244	13.144	0.025	0.224		
<i>Adj R²</i>	0.96	0.98	0.97	0.90	0.84	0.31		

Notes: This table provides the estimation results for equation (1), focusing on 30-day readmission and mortality. In panel A, the outcome variables in columns 1–3 measure the logarithm of the number of unplanned readmissions for pneumonia ($\log(\text{Pneumonia Readmissions})$), heart failure ($\log(\text{Heart Failure Readmissions})$), and acute myocardial infarction (denoted AMI; $\log(\text{AMI Readmissions})$), respectively. The outcome variables in columns 4–6 measure the readmission rates for pneumonia, heart failure, and AMI, respectively. The outcome variable in column 7 is the readmission rate for all diseases ($\text{All Readmission Rate}$). The outcome variable in column 8 is a dummy variable that takes a value of 1 if the hospital's CMS-reported readmission rate for all diseases is in the worst-performing group among hospitals nationwide, and 0 otherwise. In panel B, $\log(\text{Pneumonia Mortality Num})$, $\log(\text{Heart Failure Mortality Num})$, and $\log(\text{AMI Mortality Num})$ are the logarithms of the number of pneumonia, heart failure, and AMI deaths, respectively. $\text{Pneumonia Mortality Num}$ is the number of pneumonia deaths. $\text{Pneumonia Mortality Rate}$ is the mortality rate for patients treated for pneumonia. $\text{Pneumonia Mortality Worst}$ is a dummy variable that takes a value of 1 if the hospital is in the worst category in terms of pneumonia deaths relative to the national average, and 0 otherwise. $STExposed$ takes a value of 1 if at least one of hospital i 's relationship banks experienced a stress test in year $t - 1$ or earlier, and 0 otherwise. Control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Year and hospital fixed effects are included, as indicated. The mean and standard deviation for each dependent variable (denoted Y) are reported (presented nonlogged if the dependent variable is a logarithm). Standard errors are clustered at the hospital level and provided in parentheses.

increase in the readmission rate for affected hospitals. Additionally, in column 7, we find that readmission rates increase for a broader set of diagnostic groups and with a similar magnitude, which suggests that the effect is not limited to the three diagnostic groups for which we have detailed data.⁴⁶ Moreover, we note that the coefficient

⁴⁶In addition to the aforementioned three, this measure includes conditions such as chronic obstructive pulmonary disease, coronary artery bypass graft surgery, elective primary total hip arthroplasty and/or total knee arthroplasty, as well as several others.

estimates belie the magnitude of the effects, as readmission rates are extremely difficult for hospitals to reduce. To put this number in context, the Affordable Care Act, in an attempt to improve health care quality, established the Hospital Readmissions Reduction Program (HRRP) in 2010 and reduced readmission rates by 0.6 to 1 percentage point after a substantial effort.⁴⁷ Furthermore, CMS levies penalties, in the form of reductions in Medicare payments, for high unplanned readmissions relative to the national average.⁴⁸ For additional texture on the above effects, we consider the likelihood that a hospital is in the worst-performing group relative to the national rate in terms of readmissions, based on CMS criteria. Column 8 of Table 5 shows that an affected hospital is significantly more likely to be in the bottom-performing group following a credit shock. These outcomes are relevant from the hospital's perspective, as hospitals in this set receive the maximum penalty by the federal government. This finding therefore underscores the magnitude of the increase in readmission rates.

Along similar lines, we consider 30-day mortality rates and levels as a measure of patient health outcomes. A mortality is defined as a patient death within 30 days of discharge from a hospital admission (including admitted patient deaths within the hospital). One limitation of this analysis is that a significant number of observations for heart failure and AMI mortality rates are missing from our dataset, since many hospitals do not report these numbers. Moreover, mortality rates for certain diagnostic groups, such as heart failure, exhibit considerable autocorrelation, as a deterioration in quality of care for these conditions may not readily impact the mortality rate. That is, unlike 30-day readmissions, patient deaths from heart failure may take months or years to transpire and thus may not be captured within our post-period. Therefore, we focus primarily on pneumonia mortality, as we have more data for this condition, and pneumonia, unlike heart failure, is a less persistent condition and thus the measure is more likely to reflect changes in health care quality. In addition, pneumonia is a common hospital-acquired condition which hospital overcrowding can increase the spread of, and so can be especially indicative of quality of care (Rothberg et al. 2014).⁴⁹

Table 5, panel B presents the results of this analysis. Columns 1–3 show the coefficient estimates for the change in the logarithm of patient deaths from pneumonia, heart failure, and AMI, respectively. We see a significant increase in pneumonia and AMI deaths for affected hospitals and an insignificant increase for heart failure mortality levels. With respect to pneumonia, the point estimates in panel A indicate that there is a 9.8 percent increase in mortality for affected hospitals; this amounts to an additional 718 pneumonia deaths per year indirectly due to the shock to credit access across affected hospitals (or 1.4 additional pneumonia deaths per affected hospital per year). We explore pneumonia mortality further in columns 4–6 by considering the per-hospital mortality-level increase, mortality rate, and the likelihood the hospital falls in the worst-performing group of pneumonia deaths relative to

⁴⁷ See Zuckerman et al. (2016) and Gupta (2021).

⁴⁸ The benchmark has since been updated through the Cures Act to be one of five peer groups based on the proportion of the hospital's patients that are dually eligible for Medicare and full-benefit Medicaid (effective starting 2019).

⁴⁹ Rothberg et al. (2014) find that 34 percent of the pneumonia hospitalizations are due to hospital-acquired pneumonia infections.

TABLE 6—HOSPITAL CARE QUALITY: PATIENT'S PERSPECTIVE

	<i>Overall Rating</i> (1)	<i>Pain Control</i> (2)	<i>Recommend</i> (3)	<i>Cleanliness</i> (4)	<i>Doctor Communication</i> (5)	<i>Nurse Communication</i> (6)	<i>Recovery Information</i> (7)	<i>Quietness</i> (8)
$STExposed_{it-1}$	−0.008 (0.002)	−0.007 (0.001)	−0.006 (0.002)	−0.006 (0.002)	−0.006 (0.001)	−0.003 (0.001)	−0.005 (0.001)	−0.009 (0.002)
<i>Controls</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>N</i>	21,372	21,358	21,370	21,372	21,372	21,372	21,371	21,372
<i>Y mean</i>	0.702	0.703	0.710	0.726	0.809	0.783	0.853	0.600
<i>Y SD</i>	0.089	0.055	0.097	0.076	0.052	0.056	0.046	0.101
<i>Adj R²</i>	0.82	0.59	0.85	0.76	0.77	0.78	0.72	0.85

Notes: This table provides the estimation results for equation (1), focusing on hospital care quality from the patient's perspective. The outcome variables are the shares of patients that give the highest rating to questions on overall care quality (*Overall Rating*), pain control, recommendation of the hospital to similar patients (*Recommend*), cleanliness, doctor communication, nurse communication, recovery information, and quietness, respectively. *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year $t - 1$ or earlier, and 0 otherwise. Control variables include the lagged logarithm of one plus available bed days ($\log(BedDays_{it-1})$) and lagged cash holdings over total assets ($Cash/TA_{it-1}$). Year and hospital fixed effects are included, as indicated. The mean and standard deviation for each dependent variable (denoted *Y*) are reported (presented nonlogged if the dependent variable is a logarithm). Standard errors are clustered at the hospital level and provided in parentheses.

their peer group, respectively. We see a significant increase for affected hospitals across all three measures. Results for both readmissions and mortality under the propensity score-matched specification are included in online Appendix Table A.6.

Patient Satisfaction

Finally, we explore patient satisfaction information from HCAHPS as a subjective measure of care quality. These measures include survey responses from randomly chosen adult patients shortly after discharge. Table 6 shows that, across all question categories, patient satisfaction significantly declines at affected hospitals relative to unaffected hospitals. The magnitudes of reduction are also consistent across all measures. Notably, patient communication with doctors and nurses becomes significantly worse, patients are less satisfied with pain control, and patients are less likely to recommend the hospital. These results are in line with the aforementioned findings of a decrease in care quality and are consistent with the notion that the medical staff is less attentive to patients in affected hospitals.⁵⁰

Put together, the strongly consistent results across our measures (readmission, mortality, and patient satisfaction) indicate a significant decline in the quality of care and patient health outcomes among hospitals that experience a credit supply shock. This suggests that hospitals are changing their operations to boost internally generated funds at the expense of patient care quality. In Section IV, we examine in more detail the specific mechanisms through which hospitals may be doing so.

⁵⁰Results on patient satisfaction for the propensity score-matched specification are provided in panel C of online Appendix Table A.6.

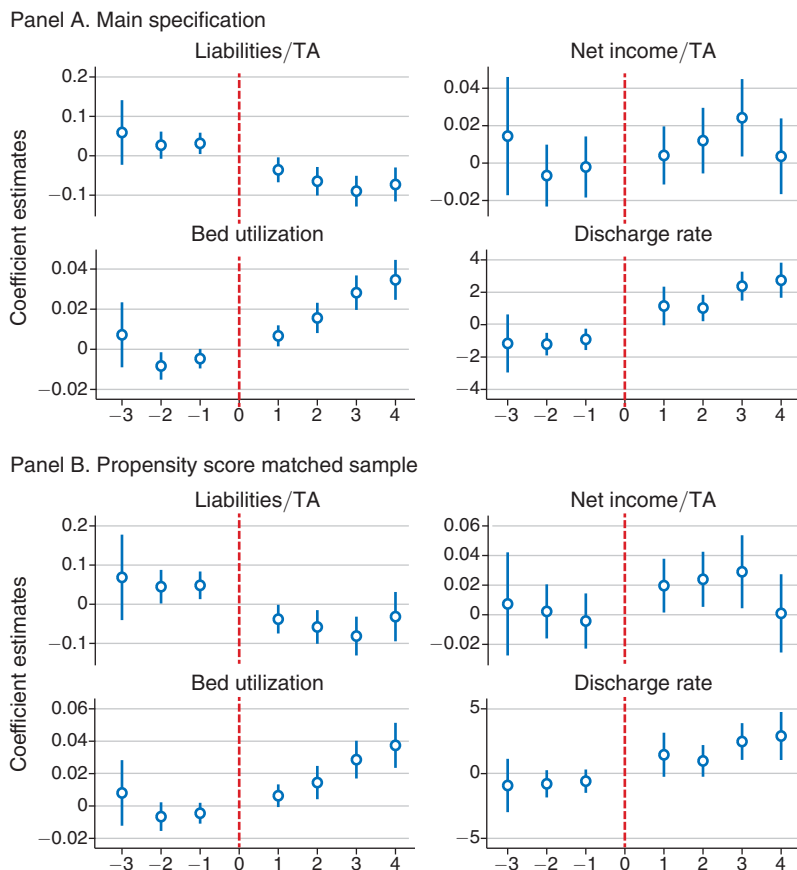


FIGURE 1. PARALLEL TRENDS: HOSPITAL FINANCIAL AND BED UTILIZATION PERFORMANCE

Notes: This figure provides parallel trends for the financial and bed utilization outcome variables by graphing estimation results for equation (3). Each coefficient represents the relative difference between the treatment and control group s years after the first exposure year (year 0). All coefficient estimates are relative to year 0. 95 percent confidence intervals are indicated by the solid lines. We plot the parallel trends in the full sample in panel A and in the propensity score matched sample in panel B.

D. Parallel Trends

The validity of our DID approach rests on the parallel trends assumption, which we now examine. Specifically, we estimate a variant of equation (1) as follows

$$(3) \quad Y_{i,t} = \alpha + \sum_{s=-3}^{-1} \beta_s \text{Exposed}_{i,t}^s + \sum_{s=1}^k \beta_s \text{Exposed}_{i,t}^s + \gamma' \mathbf{Controls}_{i,t} + \eta_t + \mu_i + \varepsilon_{i,t}.$$

In equation (3), $\text{Exposed}_{i,t}^s$ equals one if hospital i was exposed to a stress-tested bank for the first time in year $t - s$, and is equal to zero otherwise. For example, $\text{Exposed}_{i,t}^{-3}$ equals one for the year t that is three years before when hospital i 's lending banks are first stress-tested (year 0). When estimating equation (3), we omit $\text{Exposed}_{i,t}^0$, thus setting year 0 as the reference year. The interpretation of β_s is

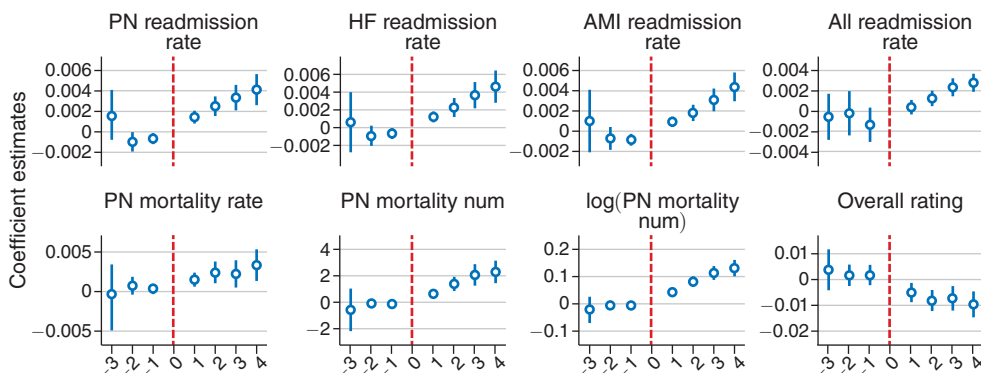


FIGURE 2. PARALLEL TRENDS: HEALTH CARE QUALITY

Notes: This figure provides parallel trends for the readmission, mortality, and survey outcome variables by graphing estimation results for equation (3). Each coefficient represents the relative difference between the treatment and control group s years after the first exposure year (year 0). PN is pneumonia, HF is heart failure, and AMI is acute myocardial infarction. All coefficient estimates are relative to year 0. 95 percent confidence intervals are indicated by the solid lines.

that it captures the relative difference between the treatment and control groups in each year, relative to the reference year 0. The parameter k denotes the maximum post-treatment year; k equals five for variables that are available in 2017, and is equal to four otherwise.

Figures 1 and 2 provide parallel trends for our main outcomes related to hospital financials, bed utilization, readmission rates, mortality, and patient satisfaction. In Figure 1, we show parallel trends for both our main sample (panel A) and our propensity score-matched sample (panel B); we include parallel trends for the other main sample and propensity score-matched sample outcomes in online Appendix Figures A.1 and A.2. For all of the variables, there are no significant pre-trends prior to the treatment year. However, after the treatment year, the variables all move immediately in the documented directions. This provides evidence that the parallel trends assumption holds in our setting. We provide additional validation tests of the parallel trends in online Appendix B.⁵¹

A potential concern in examining parallel trends in our setting stems from the fact that we utilize a DID specification with a staggered treatment and two-way fixed effects. A recent econometrics literature (Callaway and Sant'Anna 2021) has argued that the coefficients of equation (3) can potentially be hard to interpret due to variation of the control groups over time. Moreover, equation (3) tests the parallel trends assumption after conditioning on observed covariates, which can be possibly affected by the shock as well. As a robustness check, we follow the

⁵¹To more formally test the parallel trends assumption, we first perform an F -test for each of our outcome variables, which tests whether the differences between treated and control hospitals significantly vary from each other in the years leading up to DFAST implementation. For the few outcomes where there is a significant difference between the pre-period parallel trend coefficients, we examine the validity of our inferences by following the approach of Rambachan and Roth (2023), which tests the extent to which post-period treatment coefficients deviate from a trend based on the pre-period (i.e., using an expected counterfactual trend in the absence of treatment effects). We show that, even under conservative assumptions, our inferences remain valid for our main outcomes when accounting for potential pre-trends.

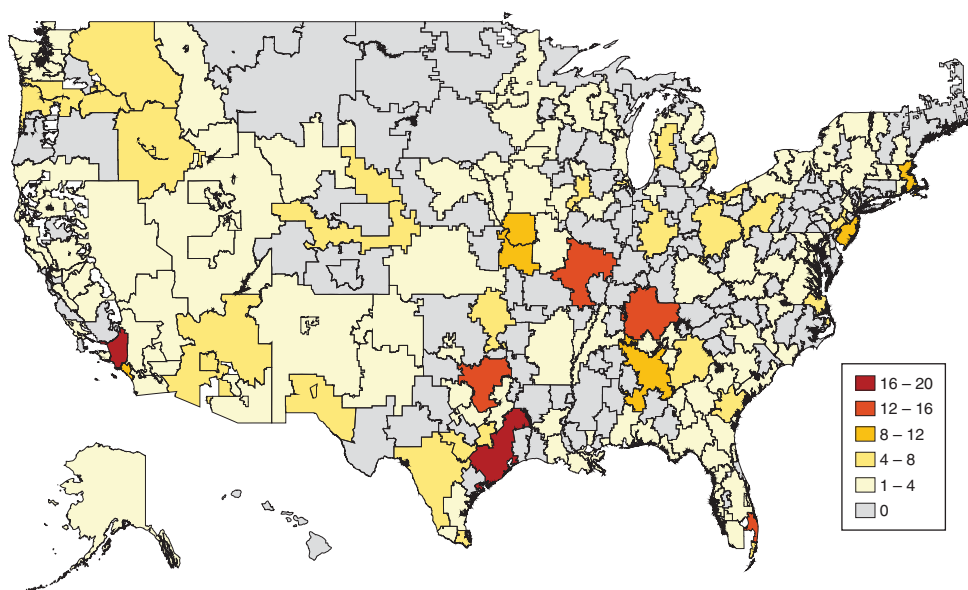


FIGURE 3. GEOGRAPHICAL DISTRIBUTION OF HOSPITALS EXPOSED TO THE STRESS TESTS

Notes: This figure shows the number of hospitals exposed to bank stress tests in different hospital referral regions (HRRs). Gray areas represent the control group.

method developed in Callaway and Sant’Anna (2021) to plot the parallel trends (see online Appendix B for details). This method does not condition on observed covariates and also provides treatment effects relative to control hospitals that are either never-treated or not-yet-treated, and thus circumvents issues related to interpreting two-way fixed effects DID regressions in a causal manner. We provide these graphs in online Appendix B; the main parallel trend plots are very similar to the previous graphs, regardless of whether we include never-treated or not-yet-treated observations as the control group for earlier events. In addition, in Table B.1 of online Appendix B, we provide the corresponding regression coefficient estimates of average treatment effects for the treated (ATTs), with never-treated hospitals as the control group, using the procedure of Callaway and Sant’Anna (2021). We also confirm that our main results are similar when dropping all covariates from the main specification (panel B of Table B.1).

IV. Mechanisms

We now explore the mechanisms underlying the main results presented in Section III. To this end, we utilize additional sources of data. First, we supplement our main data with detailed microdata from the State Inpatient Databases (SID) from the Health Care Cost and Utilization Project (HCUP). Since the SID covers the inpatient information from all payers (including private insurers), a single state typically has over 0.5 million yearly observations. To perform the regression analysis, we aggregate the encounter information at the annual hospital-DRG level (e.g., Lynk 1995; Krishnan 2001; Melnick and Keeler 2007), where the DRG indicates the

diagnosis-related group recorded for each inpatient admission, regardless of payer status.⁵² This permits us to examine effects for patient sub-groups within hospitals over time, and thus track hospital operational decisions (such as admission volume) with regard to specific patient populations. For example, all 2014 encounters of patients coded with the DRG “diabetes with complication or comorbidity” in UPMC Presbyterian Hospital are aggregated into one observation.

More specifically, we estimate the following specification at the hospital-DRG-year level:

$$(4) Y_{i,d,t} = \alpha + \beta STExposed_{i,t-1} + \gamma' \mathbf{X}_{i,t-1} + \theta' \mathbf{Z}_{i,d,t-1} + \eta_{d,t} + \eta_{i,d} + \varepsilon_{i,d,t}.$$

As in our main specification, the treatment and main explanatory variable is $STExposed_{i,t-1}$, which is equal to one if hospital i 's relationship lender experienced a stress test in year $t - 1$ or earlier, and zero otherwise. As noted previously, 15 states in the SID through our access have coverage both prior and subsequent to the stress tests, resulting in 1,448 unique hospitals, which covers about 40 percent of the sample in our main analyses in Section III. Together, we collect roughly 1.75 million hospital-DRG-year observations, with 214,521 of these observations exposed to the treatment.⁵³ The vector $\mathbf{X}_{i,t-1}$ represents the set of hospital-level control variables included in specification (1), while $\mathbf{Z}_{i,d,t-1}$ captures patient characteristics in DRG d at hospital i in a given year, including (lagged) average patient age and percentages of patients under each gender and race category provided by the SID. The parameter $\eta_{d,t}$ represents DRG-year fixed effects, which control for any time-varying shocks that broadly affect particular patient populations across all hospitals. For example, CMS periodically adjusts the service intensity weights across DRGs, which may potentially influence treatment behavior and care in different DRGs. Likewise, $\eta_{i,d}$ represents hospital-DRG fixed effects, which control for persistent characteristics of patients in DRGs for a given hospital. Our specification therefore allows us to examine changes in hospital decisions pertaining to specific patient populations following the credit shock, conditional on fixed characteristics related to the patient population within the hospital and any time-varying changes to the patient population across hospitals.

In addition to the HCUP data, we also utilize hospital-level data for the process of care scores from the CMS Hospital Compare dataset.

A. Mechanisms: Financial and Operating Decisions

To better understand the changes in hospital financial and operating decisions and quality of care presented in Section IIIB, we further investigate revenues and

⁵²Most acute care hospital inpatient stays are reimbursed prospectively on a per discharge basis based on the patient's DRG (Cooper et al. 2019). For example, the formula used to calculate payment multiplies an individual hospital's payment rate by the relative weights of the DRG. The base payment rate is adjusted based on a wage index applicable and a cost of living adjustment factor to the area where the hospital is located, which tend to be absorbed by the hospital-DRG fixed effects. Each DRG weight represents the average resources required to care for cases in that particular DRG, relative to the average resources used to treat cases in all DRGs, which will be absorbed by the DRG-year fixed effects.

⁵³Our previous hospital-year level results provided in Section III also hold for the subsample of states for which we have HCUP data. These results are available upon request.

TABLE 7—HOSPITAL DRG INPATIENT ADMISSION DECISIONS

	log(<i>Admis- sions</i>)	log(<i>Physician Order Admissions</i>)	log(<i>ED Admissions</i>)	log(<i>Elective Admissions</i>)	log(<i>Clinic Admissions</i>)	log(<i>Transfer Admissions</i>)	log(<i>Admis- sions</i>)	log(<i>ED Admissions</i>)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. General admission decisions</i>								
<i>STExposed</i> _{<i>i,t-1</i>}	0.046 (0.012)	0.069 (0.037)	0.112 (0.041)	0.062 (0.021)	-0.129 (0.033)	-0.035 (0.015)	0.060 (0.015)	0.132 (0.047)
<i>Weight</i> _{<i>d,t</i>} × <i>STExposed</i> _{<i>i,t-1</i>}							-0.008 (0.003)	-0.011 (0.005)
<i>Controls</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>DRG-Year FE</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Hospital-DRG FE</i>	Y	Y	Y	Y	Y	Y	Y	Y
<i>Y mean</i>	31.756	22.811	15.900	4.241	2.704	4.198	31.756	15.900
<i>Y SD</i>	125.327	86.353	55.727	44.758	26.851	65.367	125.327	55.727
<i>N</i>	1,651,990	1,409,410	1,409,410	1,409,410	1,409,410	1,409,410	1,651,990	1,409,410
Adj <i>R</i> ²	0.92	0.85	0.85	0.77	0.71	0.80	0.92	0.85
	<i>Private</i> log(<i>Admis- sions</i>)	<i>Medicare</i> log(<i>Admis- sions</i>)	<i>Medicaid</i> log(<i>Admis- sions</i>)					
Insurance:	(1)	(2)	(3)					
<i>Panel B. Admission decisions based on insurance types</i>								
<i>STExposed</i> _{<i>i,t-1</i>}	0.040 (0.014)	0.031 (0.012)	0.024 (0.012)					
<i>Controls</i>	Y	Y	Y					
<i>DRG-Year FE</i>	Y	Y	Y					
<i>Hospital-DRG FE</i>	Y	Y	Y					
<i>N</i>	1,651,990	1,651,990	1,651,990					
<i>Y mean</i>	9.825	12.941	6.606					
<i>Y SD</i>	62.097	37.934	49.371					
Adj <i>R</i> ²	0.85	0.87	0.83					

Notes: This table provides the regression results for equation (4) for DRG inpatient admission decisions. *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* - 1 or earlier, and 0 otherwise. In panel A, *Weight*_{*d,t*} is the MS-DRG relative weight of DRG *d* in year *t*. *Admissions* is the number of admissions assigned DRG *d* to hospital *i* in year *t*. *Physician Order Admissions* is the number of admissions via physician orders. *ED Admissions* is the number of admissions via physician orders that originate from emergency departments. *Elective Admissions* is the number of admissions via physician orders that are elective. *Clinic Admissions* is the number of admissions through clinics and physician centers. *Transfer Admissions* is the number of admissions transferred from other health care facilities such as (other) hospitals, SNFs, and ICFs. In panel B, we study admission decisions based on insurance types. Columns 1, 2, and 3 study the admission amount for privately insured, Medicare, and Medicaid patients, respectively. Hospital-level control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Hospital-DRG level control variables include the lagged average patient age and percentages of patients being female, white, black, and Hispanic. DRG-Year and Hospital-DRG fixed effects are included, as indicated. The mean and standard deviation for each dependent variable (denoted *Y*) are reported (presented nonlogged if the dependent variable is a logarithm). Standard errors are clustered at the hospital level and provided in parentheses.

costs using the inpatient database discussed above. We find that hospitals are able to achieve the gains in revenues and profitability by enhancing both resource utilization and cost efficiency. We discuss the results related to each of these separately below.

Resource Utilization.—As noted in Section IIIB, affected hospitals increase revenues from patients following the credit shock. To examine the potential sources of revenue generation, we consider inpatient admissions in Table 7 based on specification

4 above. Column 1 in Table 7, panel A indicates that inpatient admissions per DRG increased by 4.6 percent relative to unaffected hospitals. This finding reinforces our earlier results that hospital bed utilization and discharge rates increase for affected hospitals. Column 2 of panel A further shows that inpatient admissions arising from a physician order point of origin, which account for 73 percent of admissions, increases by 6.9 percent per DRG for affected hospitals. In columns 3 and 4, we see that this increase is driven by increases in admissions from the hospital's emergency department (ED), which account for 50.1 percent of admissions for the average hospital in our SID sample, and by increases in admissions for scheduled nonemergency (i.e., elective) procedures that require an inpatient stay (such as hip replacements). Moreover, admissions arising from inpatient transfers from clinics (column 5) or from other health care facilities (column 6) significantly decline.⁵⁴ The decline in transfer admissions may be due to decreased hospital bed availability as affected hospitals operate more towards capacity following the credit supply shock.

To better understand this increase in admissions, we interact our main treatment variable, $STExposed_{i,t-1}$, with a given DRG's relative weight in year t (obtained from CMS 2010d), $Weight_{d,t}$, and consider the change in total admissions and ED admissions in particular. We find that the coefficients on the interaction terms are negative and significant (columns 7 and 8),⁵⁵ which indicate that the increase in all admissions and those from the ED are greater for DRGs which correspond to conditions that are less complicated or less severe. This set of results is consistent with hospitals lowering the standard for admission from the ED and admitting patients with less severe conditions.⁵⁶ In online Appendix Table A.8, we similarly analyze the Case Mix Index (CMI) using our main specification (1), which is provided by CMS (2010–2016a) and represents the average severity of patient conditions (who are covered by Medicare) for a given hospital-year. In line with the aforementioned results, an affected hospital's CMI significantly decreases, suggesting a less severe patient pool after the shock (column 1 of online Appendix Table A.8). These results also help to mitigate concerns that our documented effects in Section IIIC are driven by contemporaneous declines in patient conditions.

As a falsification test, we consider heart attack and childbirth admissions, which are less gameable by hospitals and thus should not experience a similar increase in admissions.⁵⁷ We find insignificant changes under both of these conditions in online Appendix Table A.9.

⁵⁴In our sample, admissions from inpatient transfers from clinics account for 9.6 percent of admissions, while transfers from other health care facilities account for 7.8 percent of admissions. Admissions for scheduled elective procedures account for 13.8 percent of admissions. Physician-ordered admissions include both elective and ED admissions.

⁵⁵To understand the coefficient magnitudes, consider the diagnosis of diabetes in the 2016 MS-DRG system as an example. DRG 637, diabetes with major complication or comorbidity (MCC), has a relative weight of 1.3823, and DRG 639, diabetes without CC/MCC, has a weight of 0.6007. The coefficient in column 7 implies that admissions coded with DRG 637 increase by 4.9 percent for the average affected hospital, whereas inpatient admissions coded with DRG 639 increase by 5.5 percent.

⁵⁶Relatedly, in online Appendix Table A.10, we find that conditions for which ED admissions comprised a lower share of total inpatient admissions for that condition exhibited a larger increase following the shock, consistent with hospitals increasing admissions for conditions that did not often require inpatient hospitalization following an ED visit.

⁵⁷While we find an insignificant change, we note there is some evidence that childbirth admissions are potentially gameable through increasing referrals of childbirth patients from physician offices and prenatal clinics. As stated in US Department of Justice (2016): "According to the criminal information, as part of the scheme, expectant

In terms of admissions by primary payer type, the largest increase is among privately insured patients, followed by Medicare patients and lowest among Medicaid patients, as shown in panel B of Table 7. We find similar results with the CMS data as well, whereby the fraction of discharges that are Medicare patients is significantly lower for affected hospitals, and that of the Medicaid group also insignificantly drops (columns 2 and 3 of online Appendix Table A.8).

Hospitals can also increase revenues through coding inpatients to DRGs with higher reimbursement rates (i.e., “upcoding”). As noted previously, in contrast to upcoding, we find that inpatients are coded more to DRGs with *lower* relative weights (columns 7 and 8 in Table 7, panel A).⁵⁸

Cost Efficiency.—Affected hospitals also exhibit a significant increase in their profit margins, as documented in Table 4. As profit (or net income) is determined as revenues minus costs, hospitals can attempt to improve cost efficiency to boost their profitability. Similar to an increase in revenues, cost-saving measures allow the hospital to rely on their internally generated funds to finance their future operating activities rather than seeking external financing.

As noted previously, inpatient stay reimbursements are set prospectively for a fixed dollar amount based on the DRG for which the patient is assigned. This system is used for Medicare and Medicaid inpatient services, as well as for privately insured patients whose contracted insurer payment structures are based on Medicare or a fixed/prospective payment scheme. This fixed pricing structure appears to be common in private insurance contracts; Cooper et al. (2019) find that fixed payments for privately insured patients comprise about 77 percent of inpatient cases in their sample.⁵⁹ The DRG-based reimbursement accounts for the hospital resources, including medical procedures and the patient’s hospital stay, required to treat the condition under a given DRG. Reducing the number of procedures and length of stay for a given patient can therefore allow the hospital to receive the same payment at a lower cost for a given DRG.⁶⁰

To explore this mechanism, we investigate changes in the average number of procedures for a given DRG and the average length of stay per DRG through specification (4). Column 1 of Table 8, panel A, shows that the average number of procedures for inpatients per DRG significantly decreases for affected hospitals. Aggregating across DRGs within a hospital, the point estimate in column 1 implies a reduction of 408 procedures provided to admitted patients per year for a given affected hospital

mothers were in some cases told at the prenatal care clinics that Medicaid would cover the costs associated with their childbirth and the care of their newborn only if they delivered at one of the Tenet hospitals, and in other cases were simply told that they were required to deliver at one of the Tenet hospitals, leaving them with the false belief that they could not select the hospital of their choice.”

⁵⁸We also do not observe a significant change in average listed prices (i.e., charges) for privately insured patients among affected hospitals relative to unaffected hospitals, as shown in online Appendix Table A.11.

⁵⁹The remaining 23 percent of cases are estimated to be reimbursed as a share of the hospital’s listed price (charge). Of the 77 percent of inpatient cases that are based on fixed payments, Cooper et al. (2019) estimate that 74 percent of these are determined as a share of Medicare reimbursement rates.

⁶⁰We note that some hospital procedures during inpatient treatment have separate charges that can be on top of the DRG-based reimbursement. Hence, we might expect that charge per patient should decrease by the reduction in number of procedures. However, the fact that average inpatient charge per DRG is almost constant for affected hospitals (column 2 of Table A.11) implies that the main reduction in procedures are for those covered by the DRG-based payments.

TABLE 8—HOSPITAL DRG INPATIENT PROCEDURE AND STAY DECISIONS

Insurance:	All log(Avg Num Procedures) (1)	Private log(Avg Num Procedures) (2)	Medicare log(Avg Num Procedures) (3)	Medicaid log(Avg Num Procedures) (4)
<i>Panel A. Number of procedures</i>				
<i>STExposed</i> $d_{i,t-1}$	−0.018 (0.006)	−0.011 (0.007)	−0.017 (0.007)	−0.014 (0.008)
<i>Controls</i>	Y	Y	Y	Y
<i>DRG-Year</i> fixed effects	Y	Y	Y	Y
<i>Hospital-DRG</i> fixed effects	Y	Y	Y	Y
<i>N</i>	1,643,370	1,161,686	1,388,557	859,807
Adj <i>R</i> ²	0.81	0.74	0.77	0.71
Insurance:	All log(Avg Length of Stay) (1)	Private log(Avg Length of Stay) (2)	Medicare log(Avg Length of Stay) (3)	Medicaid log(Avg Length of Stay) (4)
<i>Panel B. Length of stay</i>				
<i>STExposed</i> $d_{i,t-1}$	−0.008 (0.006)	−0.012 (0.005)	−0.004 (0.006)	−0.011 (0.006)
<i>Controls</i>	Y	Y	Y	Y
<i>DRG-Year</i> fixed effects	Y	Y	Y	Y
<i>Hospital-DRG</i> fixed effects	Y	Y	Y	Y
<i>N</i>	1,651,986	1,161,686	1,388,557	859,807
Adj <i>R</i> ²	0.69	0.60	0.62	0.58

Notes: This table provides the regression results for equation (4) for the number of procedures (panel A) and the length of stay (panel B). *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* − 1 or earlier, and 0 otherwise. *Avg Num Procedures* is the average number of procedures for each case in DRG *d* at year *t*. *Avg Length of Stay* is the average length of stay per case in DRG *d* at year *t*. In both panels, columns 2, 3, and 4 consider the outcome separately for privately insured patients, Medicare patients, and Medicaid patients, respectively. Hospital-level control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Hospital-DRG level control variables include the lagged average patient age and percentages of patients that are female, white, black, and Hispanic. DRG-Year and Hospital-DRG fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

in our sample.⁶¹ When partitioning by payer type in columns 2–4, we find that the largest decrease in procedures is for Medicare patients. In panel B of Table 8, we see that average length of stay per DRG insignificantly decreases. While length of stay is insignificant when aggregated across payer types, there is a significant decline in the average length of stay among privately insured, Medicaid, and uninsured patients (the last coefficient is reported in column 3 of online Appendix Table A.12). For privately insured patients, column 2 signifies 279 fewer days for privately insured inpatient stays aggregated across DRGs for a given affected hospital per year.⁶²

⁶¹ About 32 patients are admitted per DRG for a given hospital in our sample, with each inpatient receiving on average 2 procedures. Each hospital codes about 355 DRGs for inpatient care in our sample. The 1.8 percent decrease from column 1 therefore translates to an aggregate per-hospital reduction of $64 \times 0.018 \times 355$.

⁶² The average number of privately insured patients admitted per DRG is 13.65 in our sample, while the average length of stay for these patients is 4.8 days. The average number of DRGs coded for inpatient care by a given hospital in a given year in our sample is 355. Across DRGs, the 1.2 percent decrease from column 2 translates to an aggregate hospital reduction of $13.65 \times 4.8 \times 0.012 \times 355$. Furthermore, in Section IIIB, we showed that *Bed Utilization* increased by 668 days; this prior estimate is the net effect on total patient bed days from increased inpatient admissions and decreased inpatient length of stay.

The insignificant decline in Medicare length of stay may be due to institutional features specific to Medicare. For example, Medicare requires a minimum inpatient hospital stay of two consecutive days for the admission to be covered by Medicare. Relatedly, Medicare requires a minimum hospital stay of three consecutive days for a Medicare patient to qualify for coverage of a subsequent Skilled Nursing Facility (SNF) inpatient stay or extended care services. Some nursing homes also require a maximum patient temperature prior to discharge before the patient can return to the nursing home.

By reducing the number of procedures and length of stay for patients, hospitals are able to scale back on costs while receiving similar payments through the DRG-based reimbursement from insurers for inpatient admissions, thereby boosting profitability per patient. Moreover, improved inpatient turnover in the hospital allows for greater admissions and thus increased revenues. Indeed, in our discussions with hospital executives and attending physicians, hospitals pay close attention to patient length of stay and the medical staff are sometimes urged to expeditiously discharge patients. For example, attending physicians reported to us that they receive a page when the hospital is close to capacity to indicate the necessity for available beds and faster patient turnover.

Summary.—The results above help to explain how affected hospitals achieve gains in revenues and profitability. The results indicate that hospitals primarily increase revenues by admitting more patients and operating closer to capacity. Hospitals appear to be lowering the standard of admission by admitting patients with less severe conditions. These results are also consistent with anecdotal evidence of hospital executives ramping up admissions to increase revenues (as discussed in Section I). Moreover, the largest increase in admissions is for patients with private insurers, which typically have the highest reimbursement rates. We additionally do not find evidence of increased prices or upcoding. At the same time, hospitals are able to improve cost efficiency and lower the cost of inpatient admissions by reducing the length of inpatient stays as well as scaling back on the number of procedures performed per DRG.

B. Mechanism: Quality of Care

An important question is why average quality of care declines for affected hospitals, despite these hospitals admitting more patients with less severe conditions. To investigate the mechanisms underlying this decline in care quality, we examine staffing and additional measures of attentiveness of the medical staff.

Staffing.—We examine whether affected hospitals increase members of the medical staff to accommodate the increase in admissions and greater bed utilization. The SID contains data on each inpatient's primary attending physician as well as other physicians who also provided care or treatment to the patient. The results are presented in Table 9. We examine the average number of admitted patients cared for by each physician in a given hospital-DRG-year in column 1. The results indicate that attending physicians cared for significantly more admitted patients on average for a given DRG in affected hospitals. This aligns with our aforementioned results of an

TABLE 9—HOSPITAL DRG STAFFING DECISIONS

	$\log(\text{Admissions per Physician})$ (1)	$\log(\text{Physicians per Patient})$ (2)	$\log(\text{Unique Physicians})$ (3)
$STExposed_{i,t-1}$	0.075 (0.028)	-0.035 (0.009)	-0.169 (0.151)
<i>Controls</i>	Y	Y	Y
<i>DRG-Year fixed effects</i>	Y	Y	Y
<i>Hospital-DRG fixed effects</i>	Y	Y	Y
<i>N</i>	1,255,084	1,651,990	1,651,990
<i>Adj R²</i>	0.84	0.90	0.95

Notes: This table provides the regression results for equation (4) for hospital staffing decisions. *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* - 1 or earlier, and 0 otherwise. *Admissions per Physician* is the average number of admissions attended by each physician in hospital *i*'s DRG *d* in year *t*. *Physicians per Patient* is the average number of physicians involved in each admission in hospital *i*'s DRG *d* in year *t*. *Unique Physicians* is the number of unique physicians providing care for patients in hospital *i*'s DRG *d* in year *t*. Hospital-level control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Hospital-DRG level control variables include the lagged average patient age and percentages of patients that are female, white, black, and Hispanic. DRG-Year and Hospital-DRG fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

increase in inpatient admissions, and suggests that individual physician workloads increase following the shock. Column 2 likewise considers the average number of unique physicians that provided care for an individual inpatient. We see that significantly fewer attending physicians are involved in a given inpatient admission. Finally, we consider the number of unique physicians providing care for inpatients in a given DRG. Column 3 indicates that the number of unique physicians remains the same, which implies that hospitals did not increase either their contracted or directly employed physicians to accommodate the increase in inpatient admissions.

Timely and Effective Care.—To measure attentiveness of the medical staff, we utilize process of care scores at the hospital level from the CMS Hospital Compare dataset. Our measures for timely and effective care include the frequency or speed with which patients receive the appropriate treatment or medical procedures after being admitted or upon discharge for the three conditions tracked closely by CMS (pneumonia, heart failure, and AMI). These measures thus reflect attentiveness of the medical staff in treating patients.

For AMI, we use three measures: (i) the portion of patients that receive aspirin at discharge; (ii) the portion of patients that receive percutaneous coronary intervention (PCI) within 90 minutes of arrival; and (iii) the portion of patients that receive a statin prescription at discharge.⁶³ For heart failure, we use the portion of patients that receive left ventricular systolic evaluations (LVS) upon arrival and ACE inhibitors

⁶³PCI is a nonsurgical procedure performed to improve blood flow of coronary circulation. Research evidence shows that it is preferable to intravenous thrombolysis for the treatment of AMI (Keeley et al. 2003). Statins are a class of drugs often prescribed by doctors to help lower cholesterol levels in the blood. Treatment with statins initiated within three to six months after AMI reduces mortality in patients with elevated cholesterol levels (Scandinavian Simvastatin Survival Study Group 1994; Sacks et al. 1996).

TABLE 10—HOSPITAL CARE QUALITY: TIMELY AND EFFECTIVE CARE

	<i>Aspirin</i> (1)	<i>PCI</i> (2)	<i>Statin Rx</i> (3)	<i>LVS</i> (4)	<i>ACE/ARB</i> (5)	<i>Antibiotic</i> (6)
<i>STExposed</i> _{<i>i,t-1</i>}	-0.002 (0.001)	-0.013 (0.004)	-0.005 (0.002)	-0.009 (0.001)	-0.008 (0.002)	-0.008 (0.002)
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>N</i>	9,203	6,327	6,937	14,384	11,193	14,657
<i>Adj R</i> ²	0.43	0.51	0.60	0.78	0.49	0.58

Notes: This table provides estimation results for equation (1), focusing on timely and effective care quality. The outcome variables in columns 1–3 measure the shares of acute myocardial infarction (AMI) patients receiving Aspirin at discharge (*Aspirin*), percutaneous coronary intervention within 90 minutes of arrival (*PCI*), and statin at discharge (*Statin Rx*). The outcome variables in columns 4–5 measure the shares of heart failure patients receiving: evaluation of the left ventricular systolic function (*LVS*), and angiotensin converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARB) at discharge (*ACE/ARB*). Column 6 measures the share of pneumonia patients receiving the most appropriate antibiotic (*Antibiotic*). *STExposed* takes a value of 1 if at least one of hospital *i*'s relationship banks experienced a stress test in year *t* – 1 or earlier, and 0 otherwise. Control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Year and hospital fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

or angiotensin receptor blockers (ACE/ARB) at discharge.⁶⁴ For pneumonia, we use the portion of patients that receive the most appropriate antibiotic at discharge.

The results are presented in Table 10. Columns 1 to 3 correspond to AMI treatments, columns 4 and 5 to heart failure treatments, and column 6 corresponds to a routine pneumonia treatment. The results show a significant reduction in timely and effective care (with the exception of receiving aspirin at discharge for AMI patients, which is marginally insignificant). As an example, patients are 1.3 percentage points less likely to receive a percutaneous coronary intervention (PCI) within the recommended 90 minutes of arrival to an affected hospital after a heart attack (AMI, column 2). PCI treatment within the 90-minute window is critical, as the survival likelihood drops significantly when the time to treatment exceeds 90 minutes. Indeed, every 10-minute treatment delay beyond this window results in an additional 3.3 deaths per 100 patients (Scholz et al. 2018).

Moreover, across five of the six measures, the likelihood of failing to provide correct or timely treatment increases by 0.5–1.3 percentage points for affected hospitals. This represents a 14–20 percent increase relative to the sample mean of 3.2–6.5 percent, depending on the treatment or procedure. The results suggest that the medical staff in affected hospitals become significantly less attentive following the credit supply shock.

⁶⁴Systolic dysfunction (when the left ventricle of the heart fails to contract normally and distribute enough blood into circulation) is a major cause of heart failure. In line with this, when the American College of Cardiology and the American Heart Association (ACC/AHA) issued detailed guidelines for the evaluation and management of heart failure in 1995, the primary focus was on systolic dysfunction. ACE inhibitors relax the veins and arteries to lower blood pressure and significantly improve the long-term survival rate after heart failure (Pfeffer et al. 1992). ARBs are considered a reasonable alternative to ACE inhibitors, particularly in patients with intolerance to ACE inhibitors.

Summary.—The above results help to link the changes in operational efficiency to changes in patient health outcomes and quality of care. Collectively, the results imply that physicians and the medical staff in affected hospitals have greater strains on their time and attention following the shock to credit supply. Hospital bed utilization increases through greater admissions and faster patient turnover. Meanwhile, the number of physicians in these hospitals remains unchanged, resulting in physicians admitting and providing care for more patients. This additional workload appears to negatively affect the quality of care provided for patients: admitted patients receive worse treatment and care, as evidenced by the process of care scores.⁶⁵ The decline in quality of care is eventually reflected in patient health outcomes as readmissions and mortality rates rise for affected hospitals. Hence, the results suggest a channel for how impediments to credit access indirectly translate to a negative impact on patient health outcomes and quality of care.

The results also do not necessarily imply affected hospitals were operating suboptimally prior to the credit supply shock. While hospitals seek to maximize profitability like other firms, they also have a health provision objective that may run counter to maximizing profitability. As such, hospitals optimize between profits and health provision (i.e., concerns over patient utility) in their objective function. Under tighter financial constraints, revenues collected from patients and profitability become more essential for the hospital. In turn, affected hospitals are forced to reoptimize and shift their decisions more towards revenues and profitability and away from health care quality.⁶⁶

V. Heterogeneity and Robustness Tests

We perform a number of additional analyses regarding heterogeneity in responses to further establish the channels behind the results as well as a variety of robustness tests. In presenting the results of these tests, we focus on a key subset of outcomes from the previous tables to minimize clutter; we note that the results are generally consistent across the other measures.⁶⁷ For several of the tests, we provide more details regarding the specifications and additional discussion of the results in online Appendix A.

A. Heterogeneity in Hospital Responses

We first consider heterogeneity in responses based on the hospital's reliance on bank loans. If a hospital is more dependent on bank loan financing, then the negative

⁶⁵This channel is also consistent with Silver (2021), who finds that quality of care is lower when emergency room doctors work faster due to workplace peer effects.

⁶⁶Previous studies have found that for-profit and nonprofit hospitals behave similarly in response to financial incentives and shocks (e.g., Duggan 2000), and therefore are unlikely to have substantial differences in their objective functions. We note that our results generally hold for both for-profit and nonprofit hospitals, as shown in Table 12.

⁶⁷In untabulated results, we provide parallel trend plots for the heterogeneity tests and furthermore provide validation of the inferences in the potential presence of pre-trends for these tests using the approach of Rambachan and Roth (2023), described in online Appendix B. These results are available upon request. We note that the coefficient for profit margin in the for-profit heterogeneity analysis (panel A of Table 12) does not satisfy the validation approach of Rambachan and Roth (2023). However, the coefficient for this variable from the heterogeneity regression analysis in Table 12 is also insignificant.

TABLE 11—HETEROGENEITY ACROSS BANK LOAN RELIANCE

	<i>Profit Margin</i> (1)	<i>Bed Utilization</i> (2)	<i>Discharge Rate</i> (3)	<i>All Readmission Rate</i> (4)	<i>Antibiotic</i> (5)	<i>Overall Rating</i> (6)
$STExposed_{i,t-1}$	0.004 (0.010)	0.011 (0.006)	0.693 (0.549)	0.001 (0.001)	−0.003 (0.003)	−0.004 (0.003)
$HighReliance_i \times STExposed_{i,t-1}$	0.010 (0.010)	0.025 (0.007)	2.665 (0.721)	0.003 (0.001)	−0.010 (0.004)	−0.004 (0.003)
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>N</i>	23,804	23,269	23,267	17,694	15,126	21,372
<i>Adj R</i> ²	0.21	0.94	0.80	0.67	0.58	0.82

Notes: This table provides estimation results when interacting the treatment variable with the hospital's reliance on bank loans. We define *reliance* as a hospital's non-matured loan amount over its total income. *HighReliance_i* takes a value of 1 if hospital *i*'s *reliance* in the year before the credit supply shock for hospital *i* is above-median, and 0 otherwise. *Profit Margin* is defined as $(Income - Cost)/Income$. *Bed Utilization* is the average daily fraction of hospital beds that are occupied. *Discharge Rate* is inpatient discharges over total bed days. *All Readmission Rate* is the readmission rate for all diseases. *Antibiotic* measures the share of pneumonia patients receiving the most appropriate antibiotic. *Overall Rating* is the share of patients that give the highest rating to questions on overall care quality. Control variables include the lagged logarithm of one plus available bed days ($\log(BedDays_{i,t-1})$), and lagged cash holdings over total assets ($Cash/TA_{i,t-1}$). Year and hospital fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

credit shock induced by stress tests should be more severe. To explore this, we first calculate each hospital's bank loan reliance, which we define to be the hospital's (non-matured) loan amount divided by its total income, both measured in the year prior to the hospital's credit supply shock. We then consider a specification where we interact our main treatment variable, $STExposed_{i,t-1}$, with $HighReliance_i$, which is an indicator that takes the value of one if hospital *i*'s pre-shock loan reliance is above-median, and zero otherwise. The results are provided in Table 11 and show a consistent pattern of stronger effects for the affected hospitals that are more reliant on loan financing.⁶⁸

Another important potential source of heterogeneity in responses is through ownership status of the hospital, such as whether the hospital is for-profit or nonprofit. To examine this, we consider a similar specification as above except we interact our main treatment variable with $ForProfit_i$, which is an indicator variable equal to one if hospital *i* is a for-profit hospital, and zero otherwise. The results of this test, which are reported in panel A of Table 12, show that affected for-profit hospitals generally have a stronger response than affected nonprofit hospitals in terms of both operational decisions and declines in quality of care. Interestingly, for-profits exhibit a lower change in their profit margins following the tightened credit constraints. This result is consistent with for-profit hospitals operating more efficiently financially or placing a greater emphasis on profits over health provision prior to the shock, thereby having less room for improvement on that dimension.

⁶⁸The parallel trends figures for the above-median bank loan reliance subgroup are very similar to those of our main sample presented in Section IIID (available upon request).

TABLE 12—HETEROGENEITY ACROSS FOR-PROFIT AND NONPROFIT HOSPITALS AND FOR-PROFIT RESTRICTED SAMPLE

	<i>Profit Margin</i>	<i>Bed Utilization</i>	<i>Discharge Rate</i>	<i>All Readmission Rate</i>	<i>Antibiotic</i>	<i>Overall Rating</i>
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Heterogeneity in main sample</i>						
$STExposed_{i,t-1}$	0.016 (0.007)	0.013 (0.005)	1.409 (0.527)	0.001 (0.000)	−0.004 (0.003)	−0.004 (0.002)
$ForProfit_i \times STExposed_{i,t-1}$	−0.011 (0.010)	0.019 (0.007)	1.970 (0.741)	0.002 (0.001)	−0.006 (0.004)	−0.008 (0.003)
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>N</i>	23,765	23,244	23,239	17,694	15,126	21,372
<i>Adj R²</i>	0.21	0.94	0.80	0.67	0.58	0.82
<i>Panel B. For-profit sample</i>						
$STExposed_{i,t-1}$	−0.003 (0.019)	0.036 (0.006)	3.901 (0.844)	0.003 (0.001)	−0.003 (0.004)	−0.007 (0.003)
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>Hospital fixed effects</i>	Y	Y	Y	Y	Y	Y
<i>N</i>	5,002	4,898	4,898	3,620	2,904	4,619
<i>Adj R²</i>	0.21	0.91	0.58	0.68	0.60	0.85

Notes: This table provides estimation results when interacting the treatment variable with a hospital's tax status. Panel A estimates the interaction results in the main sample. $ForProfit_i$ takes a value of 1 if hospital i is a for-profit hospital, and 0 otherwise. *Profit Margin* is defined as $(Income - Cost)/Income$. *Bed Utilization* is the average daily fraction of hospital beds that are occupied. *Discharge Rate* is inpatient discharges over total bed days. *All Readmission Rate* is the readmission rate for all diseases. *Antibiotic* measures the share of pneumonia patients receiving the most appropriate antibiotic. *Overall Rating* is the share of patients that give the highest rating to questions on overall care quality. Panel B estimates the main specification when restricting the sample to for-profit hospitals only. Control variables include the lagged logarithm of one plus available bed days ($\log(BedDays_{i,t-1})$), and lagged cash holdings over total assets ($Cash/TA_{i,t-1}$). Year and hospital fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

In additional analyses, we find that hospitals with greater market power (Table A.13) or lower cash holdings prior to the shock (Table A.14) exhibit a stronger response to the tightened credit constraints. The latter result suggests greater cash reserves can help to mitigate the impact of the credit supply shock as hospitals with greater cash balances can rely more on internal reserves, thus incurring a lower cost to the stress tests. As we discuss in the following section, this result has potential policy implications. Finally, we do not find significant differences based on hospital location (rural versus urban) or whether the hospital belongs to a large system. Additional details regarding these tests are provided in Section A.1 of the online Appendix.

B. Treatment Heterogeneity

To further validate that our results are driven by a credit supply channel, we explore heterogeneity in affected hospitals' treatment exposure to bank stress tests. In particular, if the credit supply channel is at play, we would expect our results to be stronger for affected hospitals borrowing from banks that are more affected by stress tests. To examine this, we exploit the fact that lenders vary in their stress test performance.

TABLE 13—HETEROGENEITY ACROSS STRESS-TESTED BANKS

	<i>Profit Margin</i> (1)	<i>Bed Utilization</i> (2)	<i>Discharge Rate</i> (3)	<i>All Readmission Rate</i> (4)	<i>Antibiotic</i> (5)	<i>Overall Rating</i> (6)
<i>CloseExposed</i> _{<i>i,t-1</i>}	0.011 (0.006)	0.024 (0.004)	2.100 (0.440)	0.001 (0.000)	−0.009 (0.003)	−0.009 (0.002)
<i>FarExposed</i> _{<i>i,t-1</i>}	−0.002 (0.008)	0.013 (0.005)	0.772 (0.480)	0.002 (0.000)	−0.003 (0.004)	0.001 (0.002)
<i>Controls</i>	Y	Y	Y	Y	Y	Y
<i>Year</i> fixed effects	Y	Y	Y	Y	Y	Y
<i>Hospital</i> fixed effects	Y	Y	Y	Y	Y	Y
<i>N</i>	23,804	23,269	23,267	17,694	15,126	21,372
<i>Adj R</i> ²	0.21	0.94	0.80	0.67	0.58	0.82

Notes: This table provides estimation results when splitting the treatment group by the lending bank's stress test performance. Following Cortés et al. (2020), we define the minimum stress-test distance (*msd*) for banks as

$$msd = \min\{\text{Tier 1 capital} - 6\%, \text{Risk-based capital} - 8\%, \text{Stressed leverage} - 4\%\}.$$

For each treated hospital *i*, we calculate the average *msd* for all of its tested lenders, weighted by the loan amount. *CloseExposed*_{*i,t-1*} (*FarExposed*_{*i,t-1*}) takes a value of 1 if hospital *i* was exposed in year *t* − 1 or earlier and the average *msd* of its tested lenders is below (above) median, and 0 otherwise. *Profit Margin* is defined as (*Income* − *Cost*)/*Income*. *Bed Utilization* is the average daily fraction of hospital beds that are occupied. *Discharge Rate* is inpatient discharges over total bed days. *All Readmission Rate* is the readmission rate for all diseases. *Antibiotic* measures the share of pneumonia patients receiving the most appropriate antibiotic. *Overall Rating* is the share of patients that give the highest rating to questions on overall care quality. Control variables include the lagged logarithm of one plus available bed days ($\log(\text{BedDays}_{i,t-1})$) and lagged cash holdings over total assets ($\text{Cash}/\text{TA}_{i,t-1}$). Year and hospital fixed effects are included, as indicated. Standard errors are clustered at the hospital level and provided in parentheses.

Banks that are closer to the regulatory minimum face greater regulatory pressure and therefore tend to reduce their credit supply more, thus generating greater financial pressure for the hospitals they lend to. We define *CloseExposed*_{*i,t-1*} (resp. *FarExposed*_{*i,t-1*}) to take a value of one if hospital *i* was exposed to a stress-tested bank in year *t* − 1 or earlier and the hospital's lenders had a below-median (resp. above-median) average distance between stress test outcomes and the regulatory minimum thresholds, and zero otherwise. Table 13 shows that the baseline effects are centered around the hospitals that are exposed to stress tests through banks closer to the threshold. We additionally find treated hospitals with a greater portion of their total loans from stress-tested banks exhibit stronger responses to the shock (Table A.18); we refer readers to online Appendix A.1 for details.

C. Robustness

In addition to the propensity score-matched analysis discussed in Section III, we analyze a number of robustness tests. We briefly summarize the results here and provide considerable more detail and discussion in online Appendix A.2.

To mitigate concerns that our results are driven by sample composition, we restrict the sample in a variety of ways and run the main specifications on the resulting subsamples. Specifically, we separately consider subsamples of only for-profit hospitals (Table 12, panel B), hospitals that are part of smaller systems with no more than five

members (Table A.20), treatment hospitals that took out new loans following exposure to the stress tests (Table A.21), and hospitals that borrowed from commercial banks (Table A.22). Across the various subsample analyses, we find results similar to that of our main analysis. Additional robustness tests include controlling for regional differences by including hospital referral region (HRR) by time fixed effects and a placebo test that considers the responses by untreated hospitals that are within the same city as treated hospitals. Finally, we provide more detail and analysis regarding the timing of the DFAST and other programs implemented by the Federal Reserve.

VI. Concluding Remarks

This paper explores the effect of credit supply shocks on hospitals. We utilize variation in stress tests conducted on banks and examine outcomes for the hospitals that these tested banks lend to. We find evidence that hospitals experience more expensive credit and reduce the amount of debt they utilize after the banks which they have relationships with undergo stress tests. In response to this negative credit supply shock, affected hospitals engage in revenue-increasing and cost-cutting actions, thereby increasing revenues and profitability. However, we also find hospitals deliver lower-quality care to patients in response to the tightened credit constraints. In particular, we find that hospitals experience a significant increase in readmission and mortality rates for major conditions and a reduction in patient satisfaction measures. The decrease in quality of care appears to be driven by decreased attentiveness by physicians and the medical staff; doctors care for more patients at affected hospitals and the timeliness of procedures drops.

Overall, our results suggest that hospitals, like other businesses, respond to increased financial pressure through changes in their operations, and in particular are dependent on credit markets. Moreover, our study helps to shed light on the importance of credit access to hospitals, and how impediments to this access can influence real operating decisions and ultimately the quality of care that patients receive. Consequently, an implication of the analysis is that hospital default risk and the corresponding impediments to credit access are important determinants of the quality of care that hospitals provide. Regulatory or public policy intervention may help to limit the impact of hospital default risk on quality of care. One such proposal is for the federal government to provide subsidized loans to hospitals—allowing hospitals another channel to access external financing—so that access to credit (and the uncertainties associated with it) has a less deleterious effect on hospital operating decisions. The federal government currently does this in some form with the banking and housing sectors (with government-backed mortgages). While access to the municipal bond market partially serves this purpose, health care bond issuance is generally specific to funding infrastructure construction or other targeted projects and investments.

Another implication of our analysis is that higher levels of capital reserves help to mitigate the effects of the credit supply shock, as shown in our heterogeneity tests. As such, a requirement that hospitals maintain capital buffers can allow hospitals to remain well-capitalized when credit access tightens and could therefore prevent hospitals from altering operating decisions in response. Likewise, as high default risk exacerbates impediments to credit access, another potential policy prescription is to regulate hospital financial leverage, thereby lowering default risk. Our study

can thus help to inform potential regulations or policies regarding hospital credit access to ultimately improve the quality of care that patients receive. Furthermore, the implications of our analysis can help to mitigate the deleterious effects of other profit shocks that extend beyond credit access, such as the immediate aftermath of the COVID event, which critically impacted hospital finances.

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